

Master Thesis

Author Profiling Of Hate Speech Spreaders

Manar Attar

Supervisor Ilia Markov
2nd reader Isa Maks

a thesis submitted in fulfillment of the requirements
for the degree of

MA Linguistics
(Language and AI)

Vrije Universiteit Amsterdam

Computational Linguistics and Text-Mining Lab
Department of Language and Communication
Faculty of Humanities

Date June 2026
Student number 2631465
Word count 8,832

Abstract

Author profiling has been studied widely on general social media, but its behaviour on hate speech, a narrower and more aggressive register, is less well understood. This thesis compares two approaches to age and gender identification on hate speech: zero-shot LLMs, and encoder models fine-tuned on PAN14, a general social media corpus. Both are evaluated on the English portion of LiLaH-HAG, a set of Facebook hate speech comments annotated with the age group and gender of their authors. The better approach depends on the attribute. Zero-shot LLMs reach a macro F1 of 0.515 on gender identification against 0.488 for the best fine-tuned encoder, while encoders do better on age identification (0.310 against 0.211). Neither result is high enough to support reliable use in practice, and the oldest age group (66+) is almost never identified correctly by any model. Error analysis points to the narrow range of topics in hate speech as a likely factor: vocabulary overlaps heavily across demographic groups, leaving little room for stylistic separation. Whether this fully explains the low performance, or whether other factors are involved, remains an open question. The findings suggest that neither approach in its current form is reliable enough for author profiling of hate speech authors.

Declaration of Authorship

I, Manar Attar, declare that this thesis, titled *Author Profiling Of Hate Speech Spreaders* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: June 2026



Signed:

List of Figures

4.1 The demographic prompt selected for all final LLM experiments. Gender and age are predicted jointly in a single API call. The prompt is truncated here for readability; the full text including output rules is provided in Appendix A. 15

6.1 Gender confusion matrices BERT and HateBERT. 26
6.2 Gender confusion matrix LLaMA-3.1-8b. 27
6.3 Age confusion matrix BERT. 28
6.4 Age confusion matrix LLaMA-3.1-8b. 28
6.5 Age prediction macro F1 by text length bucket for BERT and LLaMA-3.1-8b. 34

7.1 Word clouds by gender. Size reflects mean TF-IDF weight. 38
7.2 Word clouds by age group. Size reflects mean TF-IDF weight. 39

B.1 Age prediction confusion matrices for all five models. 54
B.2 Gender identification confusion matrices for all five models. 56

List of Tables

3.1 Label distribution in the LiLaH-HAG English test set ($n = 619$). 10
3.2 Text length (words) in the LiLaH-HAG English test set by age group. . 10
3.3 Label distribution in the PAN14 training set after preprocessing ($n =$ 11,906). 11

5.1 BERT gender identification ($n = 619$). 18
5.2 HateBERT gender identification ($n = 619$). 19
5.3 RoBERTa gender identification ($n = 619$). 19
5.4 LLaMA-3.1-8b gender identification ($n = 619$). 19
5.5 Qwen3-32B gender identification ($n = 619$). 19
5.6 Gender identification: all model comparison on LiLaH-HAG ($n = 619$). Sorted by macro F1 within each group. Best result per group in bold. . 20
5.7 BERT age identification ($n = 619$). 21
5.8 HateBERT age identification ($n = 619$). 21
5.9 RoBERTa age identification ($n = 619$). 21
5.10 LLaMA-3.1-8b age identification ($n = 619$). 21
5.11 Qwen3-32B age identification ($n = 619$). 22
5.12 Age identification: cross-model comparison on LiLaH-HAG (Markov et al., 2022). Sorted by macro F1 within each group. Best macro F1 per group in bold. 22
5.13 CroSloEngual BERT gender identification on LiLaH-HAG Slovene hate speech subset ($n = 619$). 23
5.14 LLaMA-3.1-8b gender identification on LiLaH-HAG Slovene ($n = 619$). 24
5.15 Qwen3-32B gender identification on LiLaH Slovene ($n = 619$). 24
5.16 Gender identification: cross-model comparison on LiLaH-HAG Slovene. LLaMA-3.1-8b EN shown as cross-lingual reference only. 24
5.17 Macro F1 for all models on both tasks (LiLaH EN, $n = 619$). Best result per task in bold. 25

6.1 Sample of gender false positives: female authors predicted as male by LLaMA-3.1-8b. 29

6.2 Sample of gender false negatives: male authors predicted as female by LLaMA-3.1-8b. 30
6.3 BERT age false positives: true age $\neq$ 0–25, predicted 0–25. Sorted by word count. 31
6.4 Sample of adjacent-class age errors by BERT. 33
6.5 Age prediction macro F1 by text length bucket. 35

7.1 Most distinctive vocabulary by gender (TF-IDF difference score). . . . 38
7.2 Most characteristic vocabulary by age group (mean TF-IDF weight). . 39

Contents

Abstract i

Declaration of Authorship ii

List of Figures iii

List of Tables v

1 Introduction 1

2 Related Work 4

2.1 Age Identification 4

2.2 Gender Identification 5

2.3 Author Profiling of Authors of Hate Speech 7

3 Data 9

3.1 LiLaH-HAG (Test Set) 9

3.2 PAN14 Author Profiling Corpus (Training Set) 10

3.3 Domain Difference 11

4 Methods 13

4.1 Experimental Setup 13

4.2 Zero-Shot LLM Classification 13

4.3 Fine-Tuned Encoder Models 15

5 Results 18

5.1 Gender Identification 18

5.2 Age Identification 20

5.3 Extension to Slovene 23

5.4 Summary 25

6 Error Analysis 26

6.1 Confusion Matrix Analysis 26

6.2 Gender Identification: False Positives and False Negatives 29
6.3 Age Identification: Error Patterns 31
6.4 Text Length and Prediction Accuracy 34
6.5 Summary 35

7 Discussion **36**
7.1 A Tale of Two Tasks 36
7.2 When the Domain Erases Demographic Signal 37
7.3 The Role of Morphological Cues: Evidence from Slovene 40
7.4 Limitations 41

8 Conclusion **43**
8.1 Summary of Findings 43
8.2 Future Work 44

Appendix **48**

A Prompt Templates **49**
A.1 Main Prompt Variants 49

B Full Confusion Matrices **53**
B.1 Age Prediction — All Models 54
B.2 Gender Identification — All Models 56

C Data Statements **57**
C.1 LiLaH-HAG 57
C.2 PAN14 Author Profiling Corpus 59

Chapter 1

Introduction

Hate speech online is not anonymous in the way people often assume. Someone wrote it someone with an age, a gender, a way of expressing themselves that leaves traces in the text. Author profiling is the task of reading those traces: inferring demographic characteristics from writing style rather than from identity. It is, in this sense fundamentally stylometric, the demographic signal lies not in what a person says but in how they say it, encoded in vocabulary choice, sentence structure, hedging patterns, and generational language markers.

Most author profiling research has focused on general social media Twitter posts, blog entries, product reviews. Hate speech operates in a different register topically constrained, aggressive in tone, and produced by authors who are often intentionally anonymous. Whether profiling methods developed for ordinary social media hold up when applied to hateful comments is an open question, and one with practical implications. Knowing the likely age group and gender of authors who produce hate speech can complement existing detection systems in several ways. Hilte et al. (2023) demonstrate that hate speech authorship is not demographically uniform: the age and gender distribution of authors varies across target groups and platforms, meaning that moderation strategies designed to a generic user profile may systematically miss or misclassify content from specific demographic subgroups. Demographic profiling can inform platform-level moderation by identifying which groups are disproportionately active producers of harmful content, and it can support the design of counter-speech responses that are better targeted to the demographics actually posting that content(Doğanç and Markov, 2023) since effective counterspeech differs depending on whether it addresses a young male author or an older female one. The evaluation is conducted on the English portion of LiLaH-HAG (Markov et al., 2022), a multilingual dataset of hate speech comments collected from Facebook. Each comment is annotated with both the author's gender (male or female) and age group (0–25, 26–35, 36–65, 66–), based on information available from the authors' public Facebook profiles. No labelled hate speech data of this kind was used for training either approach: the en-

coders are fine-tuned on PAN14(Rangel et al., 2014), a general social media corpus with no hate speech focus, and the LLMs receive no task-specific training at all.

This thesis investigates that question through a direct comparison of two approaches. The first uses a large language model(LLM) in a zero-shot setting, predicting the age group and gender of comment authors directly from the text without any task-specific training. The second fine-tunes a pretrained encoder model on the PAN Author Profiling datasets social media texts annotated with demographic attributes(age and gender) and evaluates it on hate speech data.

Both approaches are tested on the English portion of LiLaH-HAG (Markov et al., 2022), a dataset of Facebook comments annotated with the age group and gender of their authors. The encoder models are fine-tuned on PAN14 (Rangel et al., 2014) a corpus of general social media text, blogs, and reviews and evaluated on hate speech, making their setup a cross-dataset test of generalisation across domains. The LLMs receive no task-specific training and are applied directly to LiLaH-HAG in a zero-shot setting, with no exposure to labelled data from either domain.

Comparing these two approaches is not just a performance benchmark. They represent different assumptions about what is needed to solve the problem. Fine-tuning assumes that labelled data from a related domain transfers to a new one; zero-shot classification assumes that a model's general language knowledge is sufficient without any task-specific examples. Which assumption holds better and whether the answer differs between Age Identification and gender identification is what this study sets out to determine.

The central research question is:

> *What is the best approach for predicting the age group and gender of authors from hateful comments: zero-shot classification using LLMs, or a pretrained encoder model fine-tuned on datasets annotated for age and gender?*

Three sub-questions structure the investigation:

SQ1 How accurately can LLMs predict author's age group and gender in a zero-shot setting?

SQ2 How accurately can a pretrained encoder model fine-tuned on annotated datasets perform the same task?

SQ3 Is the best performing approach the same for Age Identification and gender identification, or do different methods perform better for different demographic attributes?

The rest of this thesis is organised as follows. Chapter 2 reviews related work on author profiling, hate speech detection. Chapter 3 describes the datasets used.

Chapter 4 covers the methodology and experimental setup. Chapter 5 reports the results. Chapter 6 analyses and interprets the findings. Chapter 7 discusses their broader implications, and Chapter 8 provides the conclusions from this work and possible directions for future works.

Chapter 2

Related Work

Author profiling is the task of inferring demographic or personality characteristics of an author from their writing, without prior knowledge of their identity. Unlike authorship attribution which identifies *who* wrote a text author profiling asks *what kind of person* wrote it. The underlying assumption is that demographic attributes leave detectable traces in writing style: in vocabulary choice, sentence structure, degree of formality, and generational language markers (Koppel et al., 2002; Schler et al., 2006). This chapter reviews work on Age Identification and Gender Identification separately, as the two tasks differ in the nature and strength of the available linguistic signal, before discussing the specific challenges that arise when author profiling is applied to hate speech data.

2.1 Age Identification

From a ML perspective Age Identification from text is typically framed as a multi-class classification problem over broad age ranges rather than a regression over exact ages. Schler et al. (2006) established early evidence that age leaves systematic traces in writing: analysing a large collection of blog posts, they found that vocabulary breadth, part-of-speech distributions, and function word usage shift measurably across age groups, with younger authors using more informal and abbreviated language and older authors producing more formal and structurally complex text.

The PAN shared task series (Rangel et al., 2013, 2014, 2015, 2016) provided the main benchmarks for competitive evaluation. PAN 2014 (Rangel et al., 2014) introduced a multilingual dataset of social media posts annotated for gender and age, covering English, Spanish. Systems relied on supervised classification using conventional ML techniques and explored a wide range of features n-grams, character-level patterns, psycholinguistic word lists, and stylometric measures combined with machine learning classifiers, with SVMs emerging as a strong baseline.

Fine-tuned encoder models improved over classical feature-based approaches. Devlin

et al. (2019) demonstrated that contextualised representations, adapted to classification through fine-tuning on a small number of labelled examples, outperform surface-level n-gram features on most PAN-style benchmarks, particularly where demographic signal is spread across subtle stylistic patterns rather than isolated keywords. However, pretrained transformers are sensitive to domain shift: when training and test distributions differ in register or topic, performance can degrade substantially (Caselli et al., 2021).

LLMs have also been evaluated on demographic inference tasks. Ziems et al. (2024) systematically tested instruction-tuned LLMs on a broad range of computational social science problems, finding that while LLMs compete with supervised baselines on coarser tasks such as sentiment and stance detection, they remain weaker on fine-grained classification where the signal is subtle. Age Identification where the difference between classes is a gradual stylistic shift rather than a distinct content marker falls into this harder category.

A persistent challenge across Age Identification work is the uneven representation of older adults in training corpora, as this demographic is less active on the social media platforms most commonly used to construct training datasets. Classifiers trained on such data learn to treat the oldest class as near-impossible and may fail to predict it even when the test distribution contains a meaningful proportion of older authors (Rangel et al., 2015). This class imbalance problem is compounded in cross-dataset evaluation settings, where the age distribution of the training corpus may not match that of the target domain. This thesis examines whether the same failure mode occurs in a zero-shot LLM setting, where data sparsity during fine-tuning cannot be the cause pointing instead to underrepresentation in the pretraining corpus itself.

2.2 Gender Identification

Computational approaches to gender identification predate age identification by at least four years, with the former established by 2002 (Koppel et al., 2002) and the latter emerging around 2006 (Schler et al., 2006). Koppel et al. (2002) demonstrated that function words and simple syntactic features can distinguish male from female authors with reasonable accuracy, establishing that gender leaves detectable traces in writing style even without explicit content markers. The signal is stylistic rather than lexical: female-authored texts tend to use more hedging expressions, first-person pronouns, emotional vocabulary, and relational language, while male-authored texts tend toward more direct and assertive constructions (Schler et al., 2006). These markers, established across blog and social media data, form the stylometric baseline that subsequent author profiling research has built upon.

As social media became the dominant source of language data, the focus shifted

to shorter and more informal texts. Burger et al. (2011) showed that gender can be inferred from Twitter posts using lexical features and metadata, though performance dropped considerably when user identity signals were excluded, illustrating the difficulty of inferring gender from writing style alone. PAN evaluation systems achieved around 60–70% accuracy for gender identification on social media benchmarks (Rangel et al., 2014), with SVMs providing competitive baselines. Fine-tuned encoder models such as BERT (Devlin et al., 2019) subsequently improved on these figures, with top systems on later PAN tasks reaching 75–85% accuracy (Rangel et al., 2015, 2016). Zero-shot LLMs do not consistently exceed these baselines: Ziems et al. (2024) find that LLM performance on demographic classification tasks is domain-sensitive and often comparable to rather than superior to supervised models, with text-only zero-shot gender identification ranging from 60–76% depending on domain. In practice, LLMs asked to predict gender tend to produce male-biased predictions, likely reflecting imbalances in pretraining data and the difficulty of inferring gender from stylistic rather than explicit cues.

A dimension that has received limited attention in the author profiling literature is the role of the target language itself in shaping the difficulty of gender identification: specifically, whether the language provides grammatical or morphological cues that make gender easier to detect, independently of writing style. In English, gender is not grammatically encoded in verb or adjective morphology, so prediction must rely on probabilistic stylistic tendencies. In morphologically rich languages such as Slovenian, past-tense *l*-participles directly encode the speaker's gender in first-person constructions, providing an unambiguous surface signal. Škrjanec et al. (2017) demonstrated this empirically, achieving 92.6% accuracy for gender identification in Slovenian Twitter by exploiting *l*-participle morphology as a primary feature substantially outperforming methods that rely on stylistic tendencies alone, and showing that grammatically encoded gender cues are reliably detectable even in informal social media text. Consistent findings in Russian suggest that morphological richness provides a systematic advantage for gender identification across typologically distinct languages (Markov et al., 2017). The findings of Schler et al. (2006) and Koppel et al. (2002) established the stylometric basis for gender and age identification, while subsequent work showed that performance is both domain- and language-dependent. This study builds on these findings by evaluating both approaches fine-tuned encoders and zero-shot LLMs on hate speech text, a domain where genre constraints may suppress the stylometric signals that prior work relied upon.

2.3 Author Profiling of Authors of Hate Speech

The intersection of hate speech and author demographics has received growing attention since the release of annotated corpora with author metadata. Hilte et al. (2023) conducted a corpus-based demographic analysis of hate speech authors across four languages using LiLaH-HAG, finding that men produce more hateful comments than women in all four languages, and that hate speech production increases with age. These patterns are not uniform: the relationship between age, gender, and hate speech varies across languages, suggesting that socio-political context moderates who produces hateful content. Mishra et al. (2018) showed that incorporating author profile information including gender improves hate speech detection, demonstrating that demographic attributes are not only descriptively interesting but also predictively useful. Huang et al. (2020) further showed that demographic factors introduce systematic bias into hate speech classifiers, with model performance varying substantially across author age and gender groups. Together, these studies establish that the demographic profile of hate speech authors matters both for understanding the phenomenon and for building fair detection systems.

Hilte et al. (2023) provide the first systematic demographic analysis of hate speech authorship, using LiLaH-HAG (Markov et al., 2022) the same dataset used in the present study a multilingual collection of Facebook comments annotated with the age group and gender of their authors across four languages (English, Dutch, Slovenian, and Croatian). Their analysis shows that the demographic composition of hate speech communities on Facebook differs substantially from that of the general social media population: men produce more hateful comments than women in all four languages, and hate speech production increases with age, with the 36–65 group being the most active. Crucially, older age groups (36+) are considerably more prevalent among hate speech authors than in standard social media corpora, which typically skew younger. These distributional findings are directly relevant to the present study: the age and gender imbalances documented by Hilte et al. (2023) in LiLaH-HAG are the same imbalances that any profiling model must learn to predict, and the overrepresentation of older authors relative to standard training corpora such as PAN14 is a key source of difficulty for the cross-domain classification task investigated here.

A deeper challenge concerns the relationship between style and content in author profiling corpora. Wegmann et al. (2022) show that stylometric models often learn topic proxies rather than genuine demographic style signals: when training and test texts share the same topic, a model can achieve high accuracy by exploiting content-based cues rather than stylistic ones. In a topically diverse training corpus such as PAN14, style and content are partially separable; but in a topically constrained domain such as hate speech where most comments address similar targets and express

similar sentiments stylistic variation is more difficult to disentangle from content. This raises the possibility that fine-tuned encoder models in cross-domain setups are penalised not only by a change in writing style between training and test, but by the loss of topic-based shortcuts that contributed to their in-domain performance. Wegmann et al. (2022) propose representing style along independent dimensions (formality, syntactic complexity) that are explicitly content-independent; these dimensions correspond closely to the stylometric cues used in the LLM prompts employed in this study.

The stylometric literature suggests that register may be a more fundamental obstacle than training-data mismatch alone. Juola (2008) argues that genre and register exert a stronger influence on surface linguistic features than individual authorial style: when all authors write within the same genre, its conventions can override the personal demographic variation that profiling models depend on. Nini (2019) makes a complementary point from a forensic perspective, cautioning that stylometric findings established in one register should not be assumed to transfer to another, as the relationship between linguistic features and demographic variables is itself register-dependent. These accounts predict that hate speech with its constrained vocabulary, aggressive framing, and shared topical focus may suppress the individual stylistic markers of age and gender rather than leaving them intact. Whether this prediction holds in practice is an empirical question that the present study addresses.

Taken together, the literature on author profiling establishes that demographic signals are detectable from text, but that their reliability depends heavily on domain, register, and the availability of sufficient training examples for minority classes. Hate speech represents a particularly challenging case: it is a topically constrained, register-uniform genre in which the stylometric signals that profiling models rely on may be systematically suppressed. This study tests that prediction directly, comparing two approaches fine-tuned encoders and zero-shot LLMs on a hate speech corpus where both domain shift and register constraints are simultaneously operative.

Chapter 3

Data

This study uses two datasets: LiLaH-HAG (Markov et al., 2022) as the evaluation set and the PAN14 Author Profiling corpus (Rangel et al., 2014) as the training source for the fine-tuned encoder models. The evaluation set, LiLaH-HAG (Markov et al., 2022), is the only available dataset annotated for both hate speech content and the age and gender of its authors; no training data with equivalent demographic annotations exists for the hate speech domain. PAN14 (Rangel et al., 2014) is therefore used as the fine-tuning source for the encoder models a general social media corpus that provides age and gender labels at scale, at the cost of a deliberate domain shift. This cross-domain setup is central to the research design.

3.1 LiLaH-HAG (Test Set)

LiLaH-HAG (Markov et al., 2022) is a multilingual dataset of hate speech comments collected from Facebook, annotated with both the content of the hate speech (target group, severity) and the demographic profile of each author. Hate speech labels were assigned by trained independent annotators who reviewed full comment threads before classifying each comment as acceptable speech or hate speech; final labels were determined by majority vote across eight annotators, with inter-annotator agreement (Krippendorff's α) ranging from 0.41 to 0.53 across languages. Author age and gender were collected separately through direct inspection of public Facebook profiles: gender was coded as binary (male or female) and age was assigned to one of four groups (0–25, 26–35, 36–65, 66–). Users whose age or gender could not be identified with sufficient certainty were excluded from the corpus. The dataset covers four languages: English, Croatian, Dutch, and Slovenian. This thesis uses only the English portion.

The English subset contains 619 comments after filtering for complete age and gender annotations. Each comment is a short Facebook post, averaging 41.8 words in length. The label distributions are shown in Table 3.1.

Table 3.1: Label distribution in the LiLaH-HAG English test set ($n = 619$).

Task	Label	Count	%
Gender	Male (M)	445	71.9
	Female (F)	174	28.1
	Total	619	100.0
Age	0–25	34	5.5
	26–35	229	37.0
	36–65	292	47.2
	66+	64	10.3
	Total	619	100.0

Table 3.2 shows the text length distribution of the LiLaH-HAG comments by age group. Comments are short overall (mean 41.8 words, median 26 words), with considerable variance: the shortest is a single word, the longest 392 words. The 0–25 group produces the shortest texts on average (25.7 words), consistent with the casual, abbreviated writing style associated with younger authors. Female authors write longer comments on average (53.0 words) than male authors (37.4 words).

Table 3.2: Text length (words) in the LiLaH-HAG English test set by age group.

Group	Mean words	Median words
0–25	25.7	16.5
26–35	43.7	25.0
36–65	42.4	26.0
66–	40.5	29.5
Overall	**41.8**	**26.0**

Two distributional features of LiLaH-HAG are relevant to the experimental design. First, the gender distribution is heavily skewed towards male authors (71.9%), reflecting the demographic reality of who produces online hate speech. Second, the age distribution concentrates in the 26–65 range (84.2%), with the youngest (0–25, 5.5%) and oldest (66+, 10.3%) groups each forming a smaller share of the test set.

3.2 PAN14 Author Profiling Corpus (Training Set)

Since no hate speech corpus with author-level demographic annotations exists for training, the encoder models are fine-tuned on the PAN14 Author Profiling corpus (Rangel

et al., 2014) one of the largest publicly available dataset with both age group and gender labels derived from social media text. PAN14 provided annotated texts in English and Spanish; only the English portion is used here. The English portion consists of blog posts and social media texts written by authors whose demographic attributes were collected via self-report.

The dataset was preprocessed for use as a training set. Long author documents were chunked into shorter passages to match the input length constraints of transformer models, and additional review texts from the same authors were included to boost the number of training examples for the underrepresented 66+ age group. After preprocessing, the training set contains 11,906 text chunks. Table 3.3 shows the resulting label distribution.

Table 3.3: Label distribution in the PAN14 training set after preprocessing (n = 11,906).

Task	**Label**	**Count**	**%**
Gender	Male (M)	5,953	50.0
	Female (F)	5,953	50.0
	Total	11,906	100.0
Age	0–25	1,910	16.0
	26–35	3,098	26.0
	36–65	6,084	51.1
	66+	814	6.8
	Total	11,906	100.0

3.3 Domain Difference

Two properties of the train–test configuration are worth highlighting explicitly, as they directly shape the interpretation of results.

Domain gap. PAN14 consists of blog posts and general social media content averaging 261 words per chunk, while LiLaH-HAG consists of short Facebook comments averaging 41.8 words. Beyond length, the two domains differ in register: PAN14 texts are relatively considered and discursive, while LiLaH-HAG texts are often emotional outbursts with distorted spelling, coded vocabulary, and deliberate obfuscation. Models trained on PAN14 therefore face a meaningful domain shift at evaluation time.

Label distribution mismatch. The age and gender distributions differ between training and test sets. The gender distribution in PAN14 is balanced (50% male, 50%

female), while LiLaH-HAG skews toward male authors (71.9%). The age distributions also differ: the 0–25 class is more prominent in PAN14 (16.0%) than in LiLaH-HAG (5.5%), while the 26–35 class is more prevalent in LiLaH-HAG (37.0%) than in PAN14 (26.0%). These distributional differences are an inherent feature of the cross-domain setup and are discussed in relation to model performance in Chapter 5.

Chapter 4

Methods

4.1 Experimental Setup

The study compares two approaches to predicting the age group and gender of hate speech authors: zero-shot classification using LLMs, and supervised classification using pretrained encoder models fine-tuned on annotated social media data. Both approaches are evaluated on the English portion of LiLaH-HAG-HAG, described in Chapter 3.

The two approaches operate under fundamentally different assumptions. The fine-tuned encoders receive supervision from PAN14 and must generalise from that source to LiLaH-HAG-HAG a deliberate cross-domain evaluation that involves differences not only in label distribution but also in genre (blog posts and social media vs. hate speech comments), register (discursive vs. aggressive), topic (general social media content vs. immigration and identity discourse), and text length (averaging 261 words per chunk vs. 41.8 words per comment). The LLMs receive no task-specific fine-tuning; in the zero-shot condition they must infer demographic attributes from world knowledge acquired during pretraining. The aim is to understand which type of approach data-driven transfer or general-knowledge zero-shot handles the author profiling in the domain of hate speech better.

All models produce predictions for the same two tasks independently: binary gender classification (M / F) and four-class age group classification (0–25 / 26–35 / 36–65 / 66–). The two tasks are not jointly optimised; each is treated as a separate classification problem.

4.2 Zero-Shot LLM Classification

4.2.1 Models

Two instruction-tuned LLMs were evaluated via the Groq API: `llama-3.1-8b-instant`(Grattafiori et al., 2024) and `qwen/qwen3-32b`(Qwen Team, 2025). Both models were queried with

identical prompts under identical conditions. Using two models of different scale allows a partial check on whether the observed pattern is model-specific or consistent across architectures.

All inference was zero-shot: no labelled examples were included in any prompt. The models were not fine-tuned or adapted in any way.

4.2.2 Prompt Design

A preliminary experiment tested three prompt variants on a subset of 100 examples before running on the full dataset. The three variants were:

Baseline A minimal prompt listing the allowed labels and output format, with no guidance on what features to attend to.

Style A prompt instructing the model to focus on writing style cues: vocabulary, punctuation, sentence length, formality, and tone.

Demographic A prompt providing explicit demographic markers for each class: for example, characterising the 66– age group as using very formal language, old-fashioned expressions, and avoiding slang; and characterising male-authored text as more direct, less hedging, and more blunt.

The preliminary experiment selected the *demographic* prompt as the best configuration. Both age and gender were predicted in a single API call using a joint prompt, producing both labels simultaneously. The output format was as follows:

```
Gender: <M or F>
Age: <0-25 | 26-35 | 36-65 | 66->
```

Model outputs followed a fixed structured format, with gender and age labels appearing on separate lines. Labels were extracted by applying regular expressions to the raw response string, searching case-insensitively for the first match of each field. For Qwen3-32B, which produces extended chain-of-thought reasoning enclosed in `<think>...</think>` tags before the final answer, the reasoning block was stripped prior to extraction so that only the final response was parsed.

All requests used a temperature of 0.1. Input texts were truncated to 1,500 characters. The Groq API enforces a rate limit of 30 requests per minute; a 2.5-second delay between requests provided sufficient headroom.

You are a linguistic analyst specialized in demographic author profiling. Predict the gender and age group of the author based on linguistic patterns.

Demographic markers to consider:

Gender:

- M: direct assertions, less hedging, more aggressive or blunt language, fewer qualifiers
- F: hedging phrases (*I think*, *maybe*, *perhaps*), more emotional expression, more qualifiers and politeness markers

Age:

- 0–25: casual language, internet slang, abbreviations, short sentences, minimal punctuation
- 26–35: mix of formal and informal, complete sentences, references to work or adult life
- 36–65: more formal vocabulary, longer sentences, traditional expressions, less slang
- 66–: very formal language, old-fashioned expressions, avoids slang, longer and more complex sentences

Output format:

```
Gender: <M or F>
Age: <0-25 | 26-35 | 36-65 | 66->
```

Figure 4.1: The demographic prompt selected for all final LLM experiments. Gender and age are predicted jointly in a single API call. The prompt is truncated here for readability; the full text including output rules is provided in Appendix A.

4.3 Fine-Tuned Encoder Models

4.3.1 Base Models

Three pretrained encoder models were fine-tuned on the PAN14 training set:

- **BERT-base-uncased** (Devlin et al., 2019) the standard English BERT model, used as the primary baseline encoder.
- **HateBERT** (Caselli et al., 2021) BERT continued-pretrained on a large Reddit corpus of abusive language. Included to test whether domain-adapted pretraining on hateful text improves author profiling on LiLaH-HAG.
- **RoBERTa** (Liu et al., 2019) used `cardiffnlp/twitter-roberta-base`, a version of RoBERTa pretrained on tweets, and replaced cross-entropy with focal loss

($\gamma = 2$) with an additional 1.5× weight boost on the 66– class, to further address age class imbalance. Both variants have individual results tables in Chapter 5.

Gender and age were trained as separate classifiers. The gender classifier is a binary classification head (M / F); the age classifier is a four-class head (0–25 / 26–35 / 36–65 / 66–). Each base model therefore produced two fine-tuned checkpoints.

4.3.2 Training Data and Label Mapping

All encoder models were trained on `pan14_colab.csv`, a preprocessed version of the PAN14 English corpus consisting of 11,906 text chunks derived from blog and social media posts. Chunks were derived by splitting long author documents into shorter passages of at most 512 tokens, and augmenting the 66– group with additional review texts from the same authors to increase its representation.

PAN14 uses a five-class age scheme (18–24, 25–34, 35–49, 50–64, 65-xx). These were mapped to the four LiLaH-HAG age groups as follows: 18–24 → 0–25; 25–34 → 26–35; 35–49 and 50–64 → 36–65; 65-xx → 66–. The mapping collapses two PAN14 classes into the 36–65 group, which explains why 36–65 is the majority class in the training set (Table 3.3).

4.3.3 Training Configuration

The BERT and HateBERT models were fine-tuned using the Hugging Face Trainer API with the following hyperparameters: learning rate 2×10^{-5}, weight decay 0.01, batch size 8, three training epochs, and a maximum sequence length of 256 tokens. A random 10% of the training data was held out for validation, with seed 42. Class weights for the age classifier were set inversely proportional to class frequency to counter the imbalance between the small 66– group and the dominant 36–65 group.

The first RoBERTa variant used the same configuration except with five training epochs and a maximum sequence length of 128. The second variant used eight training epochs, a maximum sequence length of 256, and replaced weighted cross-entropy with focal loss. Focal loss down-weights well-classified examples during training, focusing gradient updates on hard cases in this context, the underrepresented 66– group.

4.3.4 Inference on LiLaH-HAG

Trained models were evaluated by running inference over the LiLaH-HAG test set in batches of 16. Each text was tokenised with the corresponding model's tokeniser, truncated to the maximum sequence length used during training, and passed through the classification head. The argmax of the logit vector was taken as the predicted label. No ensembling was applied; each model's predictions were evaluated independently.

All models are evaluated using a standard classification report covering precision, recall, and F1 score per class, along with macro-averaged F1 across all classes.

Chapter 5

Results

This chapter reports performance of all models on the LiLaH-HAG English test set (Markov et al., 2022). Five models are evaluated: three encoder models fine-tuned on PAN14 (Rangel et al., 2014) BERT (Devlin et al., 2019), a general-purpose baseline; HateBERT (Caselli et al., 2021), adapted to abusive language; and RoBERTa (Liu et al., 2019), tweet-pretrained with focal loss and two zero-shot LLMs, LLaMA-3.1-8b (Grattafiori et al., 2024) and Qwen3-32B (Qwen Team, 2025). Results are organised by task: gender identification first, followed by age identification. For each task, precision, recall, and F1 are reported per class alongside the macro-averaged score. The chapter closes with a cross-model summary that brings both tasks together for direct comparison.

5.1 Gender Identification

Gender identification is a binary classification task (M / F) evaluated on all five models. Results for the three fine-tuned encoders are reported in Table 5.1 (BERT), Table 5.2 (HateBERT), and Table 5.3 (RoBERTa); results for the two zero-shot LLMs follow in Table 5.4 (LLaMA-3.1-8b) and Table 5.5 (Qwen3-32B). Table 5.6 brings all five models together for direct comparison, grouped by approach.

Table 5.1: BERT gender identification ($n = 619$).

Class	**Precision**	**Recall**	**F1**
Female (F)	0.267	0.322	0.292
Male (M)	0.711	0.654	0.681
Macro	**0.489**	**0.488**	**0.487**

Table 5.2: HateBERT gender identification ($n = 619$).

Class	**Precision**	**Recall**	**F1**
Female (F)	0.294	0.764	0.425
Male (M)	0.754	0.283	0.412
Macro	**0.524**	**0.524**	**0.418**

Table 5.3: RoBERTa gender identification ($n = 619$).

Class	**Precision**	**Recall**	**F1**
Female (F)	0.279	0.414	0.333
Male (M)	0.717	0.582	0.643
Macro	**0.498**	**0.498**	**0.488**

Table 5.4: LLaMA-3.1-8b gender identification ($n = 619$).

Class	**Precision**	**Recall**	**F1**
Female (F)	0.344	0.184	0.240
Male (M)	0.730	0.863	0.791
Macro	**0.537**	**0.523**	**0.515**

Table 5.5: Qwen3-32B gender identification ($n = 619$).

Class	**Precision**	**Recall**	**F1**
Female (F)	0.333	0.036	0.065
Male (M)	0.725	0.972	0.831
Macro	**0.529**	**0.504**	**0.448**

Table 5.6: Gender identification: all model comparison on LiLaH-HAG ($n = 619$). Sorted by macro F1 within each group. Best result per group in bold.

Model	P (F)	R (F)	F1 (F)	F1 (M)	Macro F1
Fine-tuned encoder models					
RoBERTa	0.279	0.414	0.333	0.643	**0.488**
BERT	0.267	0.322	0.292	0.681	0.487
HateBERT	0.294	0.764	0.425	0.412	0.418
Zero-shot LLMs					
LLaMA-3.1-8b	0.344	0.184	0.240	0.791	**0.515**
Qwen3-32B	0.333	0.036	0.065	0.831	0.448

Considering the binary framing of the task, gender results are low overall. A trivial baseline that always predicts Male the majority class at 71.9% yields macro F1 ≈ 0.42; the best model (LLaMA, 0.515) exceeds this by 0.09 points. LLMs outperform encoders on gender identification. However, female recall differs sharply across models: HateBERT achieves the highest female recall (0.764) at the cost of near-random male recall (0.283), while Qwen3-32B achieves the highest male recall (0.972) but correctly identifies only 6 of 174 female authors (recall 0.036). LLaMA achieves the most practically useful balance, with the highest macro F1 and the best female F1 among LLMs (0.240). The male bias reflects the class distribution ($\sim$72% male in LiLaH): predicting male for all examples yields a male F1 of around 0.84. Only HateBERT overrides this tendency, overcorrecting by classifying a majority of examples as female. The potential reasons for this overall low performance are examined in Chapter 7. As an additional experiment, gender identification is also evaluated on the Slovene subset of LiLaH-HAG to gain further insight into how linguistic properties of the target language affect model performance; these results are reported in Section 5.3.

5.2 Age Identification

Age identification is a four-class classification task (0–25, 26–35, 36–65, 66–) evaluated on the same five models as gender identification. Results for the three fine-tuned encoders are reported in Table 5.7 (BERT), Table 5.8 (HateBERT), and Table 5.9 (RoBERTa), all fine-tuned on PAN14 (Rangel et al., 2014); results for the two zero-shot LLMs follow in Table 5.10 (LLaMA-3.1-8b) and Table 5.11 (Qwen3-32B). Table 5.12 brings all five models together for direct comparison, grouped by approach.

Table 5.7: BERT age identification ($n = 619$).

Class	Precision	Recall	F1
0–25	0.069	0.059	0.063
26–35	0.391	0.415	0.403
36–65	0.504	0.462	0.482
66–	0.266	0.328	0.294
Macro	**0.307**	**0.316**	**0.310**

Table 5.8: HateBERT age identification ($n = 619$).

Class	Precision	Recall	F1
0–25	0.042	0.118	0.062
26–35	0.365	0.729	0.487
36–65	0.385	0.086	0.140
66–	0.000	0.000	0.000
Macro	**0.198**	**0.233**	**0.172**

Table 5.9: RoBERTa age identification ($n = 619$).

Class	Precision	Recall	F1
0–25	0.056	0.118	0.075
26–35	0.389	0.568	0.462
36–65	0.515	0.342	0.412
66–	0.211	0.062	0.096
Macro	**0.293**	**0.273**	**0.261**

Table 5.10: LLaMA-3.1-8b age identification ($n = 619$).

Class	Precision	Recall	F1
0–25	0.060	0.412	0.105
26–35	0.439	0.157	0.232
36–65	0.497	0.517	0.507
66–	0.000	0.000	0.000
Macro	**0.249**	**0.272**	**0.211**

Table 5.11: Qwen3-32B age identification ($n = 619$).

Class	Precision	Recall	F1
0–25	0.058	0.452	0.103
26–35	0.385	0.161	0.227
36–65	0.465	0.356	0.403
66–	0.000	0.000	0.000
Macro	**0.227**	**0.242**	**0.183**

Table 5.12: Age identification: cross-model comparison on LiLaH-HAG (Markov et al., 2022). Sorted by macro F1 within each group. Best macro F1 per group in bold.

Model	F1 (0–25)	F1 (26–35)	F1 (36–65)	F1 (66–)	Macro F1
Fine-tuned encoder models					
BERT	0.063	0.403	0.482	0.294	**0.310**
RoBERTa	0.075	0.462	0.412	0.096	0.261
HateBERT	0.062	0.487	0.140	0.000	0.172
Zero-shot LLMs					
LLaMA-3.1-8b	0.105	0.232	0.507	0.000	**0.211**
Qwen3-32B	0.103	0.227	0.403	0.000	0.183

Age results are uniformly poor. A uniform random baseline across four classes yields macro F1 ≈ 0.25; BERT's best result (0.310) exceeds this by 0.06 points, and both LLMs sit within 0.04 of random. Among the encoders, BERT is the only model that predicts all four age classes with non-zero F1, achieving its strongest performance on the majority class 36–65 (F1 = 0.482) while retaining some recall on the oldest group (66–, F1 = 0.294). RoBERTa, despite focal loss training designed to address class imbalance, concentrates predictions on 26–35 and 36–65, achieving 66– F1 of only 0.096. HateBERT collapses almost entirely into the 26–35 class, producing near-zero scores on 36–65 and 66–. Among the LLMs, LLaMA-3.1-8b and Qwen3-32B both fail entirely on 66– (F1 = 0.000) and show strong but misplaced recall on 0–25, a class that accounts for only 5.5% of the test set. These patterns suggest that the difficulty is not evenly distributed: the 36–65 class is reasonably classifiable across models, while 0–25, 66–, and to a lesser extent 26–35 expose the limits of both approaches. The potential reasons for these failures are examined in Chapter 7.

5.3 Extension to Slovene

As an additional experiment, the gender classification task was extended to the Slovenian portion of LiLaH-HAG ($n = 619$ SL comments). The motivation is linguistic. Slovenian encodes grammatical gender morphologically through past-tense *l*-participles, which must agree with the subject in gender. In first-person constructions the natural form for expressing personal opinion this means the verb form itself reveals the author's gender: *sem šel* (*I went*, male) versus *sem šla* (*I went*, female); *sem rekel* (*I said*, male) versus *sem rekla* (*I said*, female) (Škrjanec et al., 2017). English has no equivalent: *I went* and *I said* are identical regardless of the speaker's gender. Any Slovenian text containing first-person past-tense clauses therefore provides a direct grammatical gender cue absent in English. This makes Slovene an ideal test case for the question motivating the experiment: whether the low English gender results are driven by the absence of such morphological signal, or by the genre and topic constraints of hate speech that would suppress stylometric signals in any language.

This experiment covers gender only, since the morphological motivation is specific to gender. The Slovene encoder is `EMBEDDIA/crosloengual-bert` (Ulčar and Robnik-Šikonja, 2020), a multilingual BERT model fine-tuned on a balanced subset of 60,000 Janes-Blog texts (Fišer et al., 2016) a large Slovenian blog corpus with gender labels derived automatically from *l*-participle morphology. Both LLMs were evaluated with a Slovene-language prompt that includes explicit *l*-participle examples (*sem šla*/*šel*). Because the training corpus, target language, and available linguistic cues all differ from the English setup, Slovene results are not directly comparable to those in Sections 5.1 and 5.2.

5.3.1 Gender Identification

Table 5.13: CroSloEngual BERT gender identification on LiLaH-HAG Slovene hate speech subset ($n = 619$).

Class	**Precision**	**Recall**	**F1**
Female (F)	0.539	0.423	0.474
Male (M)	0.703	0.791	0.744
Macro	**0.621**	**0.607**	**0.609**

Table 5.14: LLaMA-3.1-8b gender identification on LiLaH-HAG Slovene ($n = 619$).

Class	Precision	Recall	F1
Female (F)	0.472	0.498	0.484
Male (M)	0.693	0.671	0.682
Macro	**0.582**	**0.584**	**0.583**

Table 5.15: Qwen3-32B gender identification on LiLaH Slovene ($n = 619$).

Class	Precision	Recall	F1
Female (F)	0.532	0.409	0.462
Male (M)	0.700	0.793	0.743
Macro	**0.616**	**0.601**	**0.603**

Table 5.16: Gender identification: cross-model comparison on LiLaH-HAG Slovene. LLaMA-3.1-8b EN shown as cross-lingual reference only.

Model	F1 (F)	F1 (M)	Macro F1
Fine-tuned encoder			
CroSloEngual BERT	0.474	0.744	**0.609**
Zero-shot LLMs (Slovene prompt)			
Qwen3-32B	0.462	0.743	**0.603**
LLaMA-3.1-8b	0.484	0.682	0.583
Reference: best English model			
LLaMA-3.1-8b EN	0.240	0.791	0.515

CroSloEngual BERT achieves the highest gender macro F1 among Slovene models (0.609). The gap between the encoder and the best Slovene LLM (Qwen3-32B, 0.603) is only 0.6 points substantially narrower than the 2.7-point gap in English, where LLMs prefformed better. Qwen3-32B female F1 in the Slovene condition (0.462) is nearly seven times higher than in the English condition (0.065), where the model classified almost all examples as male. The Slovene morphological signal is strong enough for LLMs to close most of the encoder–LLM gap through prompting alone.

Comparing the best Slovene result (CroSloEngual BERT, 0.609) to the best English result (LLaMA-3.1-8b, 0.515), performance is 9.4 macro F1 points higher in Slovene. This gap is attributable in part to the availability of the *l*-participle gender signal, which is absent in English. However, it only partially explains the low English results:

even in Slovene, where the morphological cue is present, macro F1 does not exceed 0.609 well below ceiling. This suggests that the genre and topic constraints of hate speech expressions suppress stylometric signals regardless of language, not only in English. Evidence for this is presented in Chapter 6.

5.4 Summary

Table 5.17: Macro F1 for all models on both tasks (LiLaH EN, $n = 619$). Best result per task in bold.

Model	Gender Macro F1	Age Macro F1
Fine-tuned encoder models		
RoBERTa	**0.488**	0.261
BERT	0.487	**0.310**
HateBERT	0.418	0.172
Zero-shot LLMs		
LLaMA-3.1-8b	**0.515**	**0.211**
Qwen3-32B	0.448	0.183

The results indicate that the best performing approach is not the same for both age and gender (SQ3: *Is the best performing approach the same for age prediction and gender identification, or do different methods perform better for different attributes?*): LLaMA-3.1-8b achieves the highest gender macro F1 (0.515) and the LLM group outpreforms all encoders on gender, while BERT achieves the highest age macro F1 (0.310) and the encoder group wins the LLM group on age by approximately 10 macro F1 points. Qwen3-32B outperforms HateBERT on gender (0.448 vs. 0.418) but falls below both RoBERTa (0.488) and BERT (0.487); it falls below all encoders on age. Encoders are better considering the age task but struggles with 66+ group when trained on data that underrepresents this group.

Since zero-shot LLM performance is low, looking at best gender macro F1 of 0.515 (compared to a majority-class baseline of 0.418) and best age macro F1 of 0.211 (below the random baseline of 0.25) LLMs are not used for annotating additional training data in this work. Future work could explore whether improved LLM prompting strategies, or in-domain few-shot examples drawn from labelled LiLaH data, produce annotation quality sufficient for training set augmentation.

Chapter 6

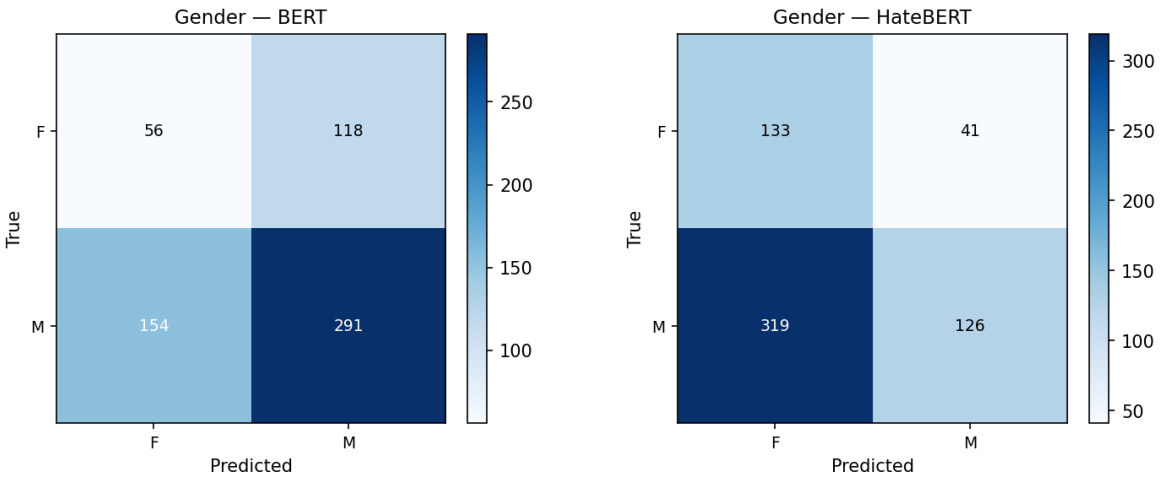
Error Analysis

This chapter examines where models succeed and fail, focusing on the best-performing model for each task: LLaMA-3.1-8b for gender identification (macro F1 = 0.515) and BERT for age identification (macro F1 = 0.310). For each task, confusion matrices situate the overall error pattern; false positives and false negatives are then examined with illustrative examples drawn directly from the test set. Interpretation in relation to the broader literature is taken up in Chapter 7.

6.1 Confusion Matrix Analysis

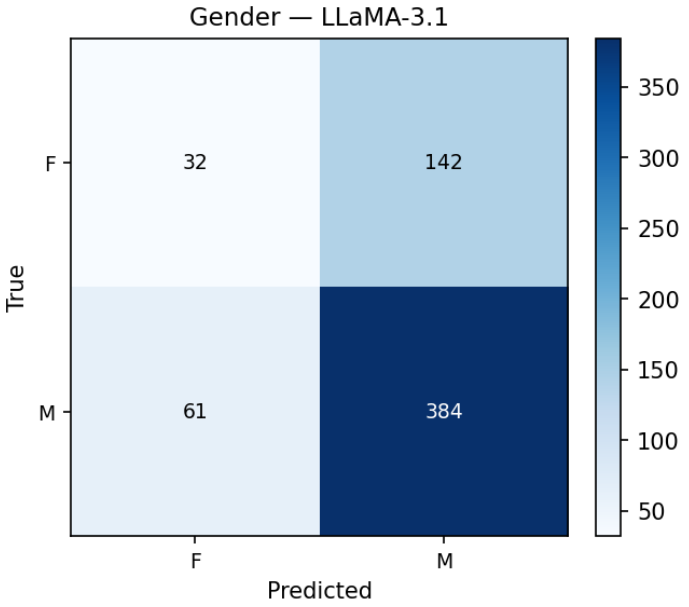
6.1.1 Gender Identification

Figures 6.1 and 6.2 show the confusion matrices for the three models discussed below; confusion matrices for all five models are provided in Appendix B.2.



BERT HateBERT

Figure 6.1: Gender confusion matrices BERT and HateBERT.



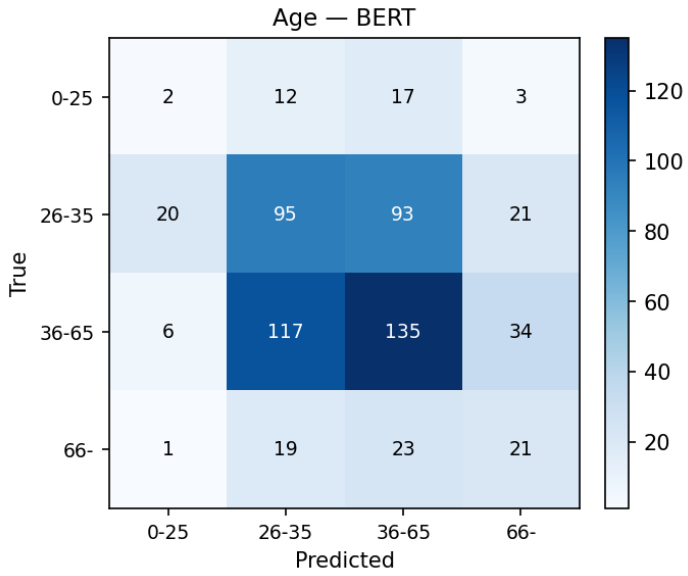
LLaMA-3.1-8b

Figure 6.2: Gender confusion matrix LLaMA-3.1-8b.

LLaMA-3.1-8b presents a strongly asymmetric error structure: 142 of 174 female examples (81.6%) are classified as male, while only 61 of 445 male examples (13.7%) are classified as female. Male recall is high (0.863) while female recall is very low (0.184). LLaMA's macro F1 advantage is primarily driven by high male recall (0.863), while its female recall (0.184) is the lowest among all models. BERT shows a more balanced pattern: it correctly identifies 56 of 174 female examples (recall 0.322) and 291 of 445 male examples (recall 0.654), but this balance does not translate to higher macro F1 because its male recall is substantially lower than LLaMA's. HateBERT is the outlier: it classifies 133 of 174 female examples correctly (recall 0.764) but at the cost of misclassifying 319 of 445 male examples as female a severe male false-negative rate that reflects the opposite bias from the other models. The cause of this reversed pattern is not investigated further here, as the focus of the FP/FN analysis is the best-performing model.

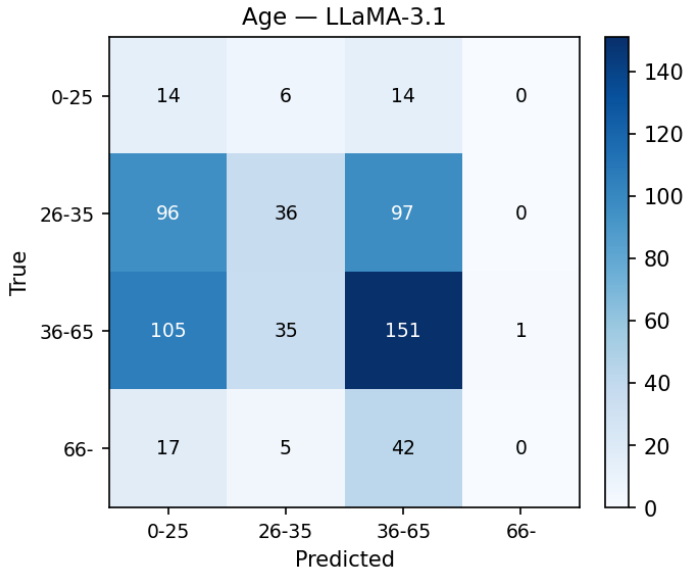
6.1.2 Age Identification

Figures 6.3 and 6.4 show the confusion matrices for the two models discussed below; confusion matrices for all five models are provided in Appendix B.1.



BERT

Figure 6.3: Age confusion matrix BERT.



LLaMA-3.1-8b

Figure 6.4: Age confusion matrix LLaMA-3.1-8b.

BERT's age confusion matrix reveals three distinct error patterns. First, the two majority classes are heavily confused with each other: 93 true 26–35 examples are predicted as 36–65, and 117 true 36–65 examples are predicted as 26–35. Second, the 0–25 class is nearly invisible: only 2 of 34 true 0–25 examples are correctly identified, with most assigned to 36–65 (17) or 26–35 (12). Third, BERT achieves the best 66– recall among the five models (21 of 64, 33%), with errors spread across adjacent classes rather than concentrated in one direction. LLaMA's confusion matrix shows a different

structure: 232 examples are predicted as 0–25 (against 34 true), achieving a 0–25 recall of 0.41 (14 of 34) through substantial over-prediction, while the 26–35 class receives only 82 predictions and the 66– class receives just one.

6.2 Gender Identification: False Positives and False Negatives

6.2.1 False Positives: Female Authors Predicted as Male

Table 6.1 shows a random sample of six of the 142 female-authored examples that LLaMA-3.1-8b classified as male. All example tables in this chapter are randomly sampled (fixed seed for reproducibility); conclusions are drawn from the full error sets, not from the displayed examples alone.

Table 6.1: Sample of gender false positives: female authors predicted as male by LLaMA-3.1-8b.

Text (truncated)	**Age**	**wc**
This site sure is pushing the Propaganda a lot lately.	36–65	10
A yank started this post. You have no problem with racist yanks, do you fascist?	36–65	15
All those refugees are selfish and primitive ppl how they will adjust to European culture.	36–65	17
I'm at a loss as to why you're sitting there obsessing about other people's sex lives. Let's discuss yours.	26–35	31
What are you backing up? Say it. Say I hate Muslims period and quit trying to defend your actions with examples of terrorism.	36–65	45
Saudi Arabia has a huge refugee camp that can literally fit hundreds of thousands if not millions of people, yet it sits empty. It's all for show.	26–35	56

The false positives are distributed across all age groups and span a wide range of text lengths (median 30 words, compared to 34 words for female-authored examples overall). A TF-IDF analysis of the false positive set against the true positive set (correctly classified female texts; noting the true positive set is small, $n = 32$) confirms a systematic topic difference: the words most distinctive for misclassified female texts are geo-political terms *culture*, *europe*, *immigrants*, *european*, *muslim*, *syria*, *country*

while correctly classified female texts are characterised by more interpersonal vocabulary: *dear*, *woman*, *children*, *respect*, *kind*. This is consistent with the model having developed an association between political and confrontational content and the male class, which may contribute to female authors who engage with these topics in the same register as male authors being misclassified. This is consistent with the genre-overrides-style argument of Wegmann et al. (2022): when genre and topic are held constant across authors, individual stylistic cues that distinguish gender are suppressed. Aggression markers (ALL-CAPS words and exclamation marks per 100 words) are higher in misclassified female texts (mean 4.81) than in correctly classified ones (mean 1.40), consistent with this interpretation, though the difference does not reach significance at $\alpha = 0.05$ ($p = 0.085$), likely due to the small number of true positives ($n = 32$).

6.2.2 False Negatives: Male Authors Predicted as Female

Table 6.2 shows a random sample of six of the 61 male-authored examples that LLaMA-3.1-8b classified as female.

Table 6.2: Sample of gender false negatives: male authors predicted as female by LLaMA-3.1-8b.

Text (truncated)	**Age**	**wc**
Learn from nature when you get confused or misplace your orientation.	0–25	11
What you mean so sad? Take them in your house if you like.	36–65	13
The BBC always makes sure there's a few strategically placed children in all its photos/video footage.	36–65	19
it is sad to see Europe or western culture end up this way. i always [been a] big admire-er of western or i would say free culture :(	36–65	26
Refugees take refuge in the nearest safe country not pick and choose and demand to be allowed to go to their country of choice with the best benefits.	36–65	28
Hopefully your local police will arrest you for hate speech. What a bigot you are.	36–65	15

The false negatives are notably shorter than the false positives (median 16 words vs. 30 words). Unlike the false positive set, which shows a clear topic signal in the TF-IDF analysis, no equivalent measurable pattern distinguishes the false negatives from

correctly classified male texts. Aggression markers (ALL-CAPS words and exclamation marks per 100 words) are slightly lower in false negative texts (mean 2.57) than in true positives (mean 4.04), and the ratio of *you*-targeting pronouns to *I*-pronouns is marginally lower (0.77 vs. 0.79), but neither difference is statistically significant ($p = 0.320$). The only consistent feature is brevity: the false negative set is shorter on average than the correctly classified male set, suggesting that very short texts may provide insufficient signal for confident male prediction, and are assigned the female class in the absence of the topic-level and aggression markers that appear to favour male predictions elsewhere. Beyond this, the 61 examples are heterogeneous and do not cluster around a single identifiable pattern.

6.3 Age Identification: Error Patterns

6.3.1 False Positives: Older Authors Predicted as 0–25

BERT predicts 0–25 for 29 examples in total, of which 27 are incorrect (false positives). Table 6.3 shows the ten shortest of the 27 false positives, sorted by word count to illustrate the brevity pattern. The median length across all 27 is 6 words, and the longest is 9 words apart from two outliers.

Table 6.3: BERT age false positives: true age ≠ 0–25, predicted 0–25. Sorted by word count.

Text	**True age**	**Gender**	**wc**
Ur still fat?	26–35	M	3
ur the demon here	26–35	M	4
ur disgusting	26–35	M	2
U need therapy	26–35	M	3
Shame on you ho to hell	36–65	M	6
1:0 for the IS then	66–	M	5
Manny Mendoza UR an idiot.	36–65	M	4
Culture is the operative word you ignorant tit x	36–65	M	9
Your an idiot dude. And everyone doesn't believe in your imaginary friend. And BTW there's no such thing as hell.	36–65	M	20
Im scared for england n us english ppl. This has to stop [. . .]	36–65	F	96

The pattern is consistent: the overwhelming majority of false positives are extremely

short, use abbreviated spelling (*ur*, *u*, *UR*, *ppl*, *n*, *BTW*), and contain direct insults. This association between abbreviated register and the 0–25 class is verifiable in the PAN14 training data itself: 0–25 authors use abbreviations at 8.2 times the rate of 36–65 authors (mean 0.275 vs. 0.034 occurrences per 100 words), and 64.4% of 0–25 texts contain at least one abbreviation compared to 51.0% of 36–65 texts. Both differences are statistically significant (Kruskal-Wallis $H = 242.7$, $p < 0.001$; $\chi^2 = 82.3$, $p < 0.001$). The most frequent abbreviated forms among 0–25 PAN14 authors are *u*, *ur*, *lol*, and *wanna* forms that overlap substantially with those appearing in the LiLaH-HAG false positive set (*ur*, *u*, *ppl*, *BTW*). BERT appears to have learned a genuine age signal from its training data; the problem is that in LiLaH, this abbreviated register characterises the hate speech genre as a whole rather than any particular age group. This is consistent with Schler et al. (2006), who found that younger bloggers use markedly more informal and abbreviated language than older bloggers, a pattern that PAN14 also a blog corpus inherits.

6.3.2 Adjacent Class Confusion: 26–35 and 36–65

The largest source of BERT age errors is bidirectional confusion between the two middle classes. Of 229 true 26–35 examples, 93 (41%) are predicted as 36–65; of 292 true 36–65 examples, 117 (40%) are predicted as 26–35. Table 6.4 shows a random sample of three examples from each direction.

Table 6.4: Sample of adjacent-class age errors by BERT.

Text (truncated)	**True**	**Pred**	**wc**
True 36–65, predicted as 26–35			
I can't understand why Europe is breaking its neck to take in these migrants you know nothing about them [. . .]	36–65	26–35	185
The EU is the biggest political con in my lifetime, the cultural & economic differences between us are huge [. . .]	36–65	26–35	69
Obama, NATO, Israel & Saudis started this mess funding the Syrian Rebels aka Al Qaeda / ISIS [. . .]	36–65	26–35	41
True 26–35, predicted as 36–65			
Excuse me? Why should we pay to keep the world's work-shy? We are like a honey-pot to the lazy and attract parasites.	26–35	36–65	23
Peter Hallam, Islam is not a race. I don't care about peoples sexuality, nationality or race, but religion is a choice [. . .]	26–35	36–65	56
Stephen Merkel acted like a teenager when the parents are away for the weekend [. . .]	26–35	36–65	70

No clear distinguishing signal separates these two groups. A pairwise Jaccard similarity analysis of the top-200 most frequent content words for each age group in LiLaH-HAG confirms this quantitatively: the 26–35 and 36–65 groups share a Jaccard similarity of 0.439, compared to 0.180 between 0–25 and 26–35, and 0.190 between 0–25 and 36–65. The two middle classes are thus more lexically similar to each other than either is to the 0–25 class. This structural overlap is even more pronounced in PAN14, where the 26–35 and 36–65 groups share a Jaccard similarity of 0.896 their top vocabularies are nearly identical. A model trained on data where these two groups share nearly all of their most frequent vocabulary may have limited basis for drawing a reliable boundary between them, regardless of domain. The bidirectional confusion at equal rates (40% in each direction) is therefore consistent with the vocabulary overlap between these groups, rather than a modelling failure specific to the cross-dataset setting. This aligns with Nguyen et al. (2013), who argue that age is better conceived as a continuum than a set of discrete classes, and that the boundaries between adjacent age groups in social media text are inherently difficult to recover from surface-level features.

6.3.3 The 0–25 Class: Near-Total Failure

Only 2 of 34 true 0–25 examples are correctly identified by BERT a per-class F1 of 0.06, the lowest for any model–class combination. The two correct predictions are:

- *"Oh my god. You're literally insane."* (M, wc=6)
- *"I wasnt born from a pizza yet I love pizza. How is this possible? Pls explain from your infinite wisdom."* (M, wc=23)

Both are informal and contain youth-associated language features hyperbolic intensifiers (*literally insane*) and playful, rhetorical phrasing even without the abbreviated spelling forms quantified in PAN14. The remaining 32 true 0–25 examples lack this surface cue and are predicted as 26–35 (12) or 36–65 (17): when the abbreviated-spelling signal is absent, the dominant topic-driven register of LiLaH-HAG hate speech provides no reliable age cue for the youngest group, and BERT assigns them to the majority classes instead. LLaMA achieves higher 0–25 recall (14 of 34, 41%) but at the cost of predicting 0–25 for 232 examples total, yielding a precision of 0.06 the model's relatively higher 0–25 recall is therefore achieved through near-indiscriminate prediction of the youngest class rather than genuine discrimination.

6.4 Text Length and Prediction Accuracy

LiLaH-HAG comments average 41.8 words, roughly six times shorter than PAN14 training chunks (mean 261 words, computed from the PAN14 training corpus). Table 6.5 and Figure 6.5 show how macro F1 varies with text length for BERT and LLaMA on the age task.

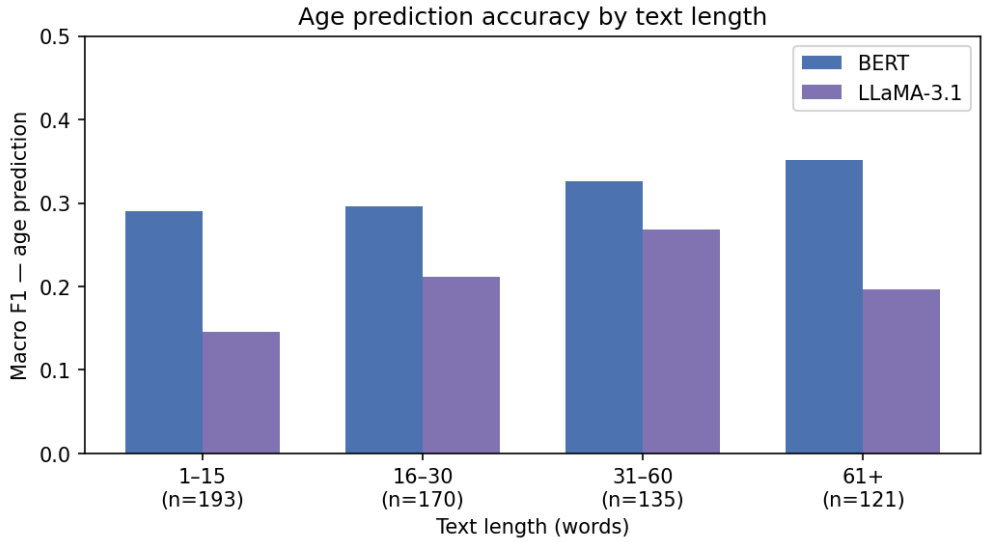


Figure 6.5: Age prediction macro F1 by text length bucket for BERT and LLaMA-3.1-8b.

Table 6.5: Age prediction macro F1 by text length bucket.

Length (words)	n	BERT macro F1	LLaMA macro F1
1–15	193	0.290	0.146
16–30	170	0.296	0.212
31–60	135	0.326	0.268
61+	121	0.352	0.197

Both models improve as text length increases from 1–15 to 31–60 words, consistent with longer texts providing more stylistic signal. BERT continues to improve in the 61+ bucket (F1 0.352), while LLaMA's performance drops (F1 0.197). One possible reading is that longer texts in LiLaH-HAG tend to be argumentative elaborations of political positions, where the topic-level content dominates any individual stylistic signature that LLaMA might otherwise use as an age cue. The pattern in the 1–15 bucket is also consistent with the false positive analysis: the majority of texts in this bucket are short, abbreviated insults, for which BERT over-predicts 0–25 and LLaMA over-predicts 0–25 even more aggressively. The broader pattern is consistent across both tasks: the male false negatives identified in Section 6.2 have a median length of 16 words, consistent with the general finding that very short texts provide less stylistic surface for demographic prediction.

6.5 Summary

Taken together, the error patterns point to two structural causes that operate across both tasks. First, the hate speech genre homogenises vocabulary across demographic groups: female authors writing about immigration and religion produce texts that are lexically very similar to male-authored texts on the same topics, and authors aged 26–35 and 36–65 share a Jaccard vocabulary similarity of 0.439 in LiLaH-HAG and 0.896 in PAN14, leaving models with no reliable lexical boundary to exploit. Second, surface features that do carry demographic signal in the training domain lose that signal in the target domain: abbreviated spelling reliably marks youth in PAN14 blog text, but in LiLaH-HAG it characterises the hate speech register as a whole. Both patterns are consistent with the same underlying explanation that genre and topic constraints appear to suppress much of the individual stylistic variation that demographic prediction depends on a finding that is taken up further in Chapter 7.

Chapter 7

Discussion

The question motivating this study was how well zero-shot LLMs and fine-tuned encoder models can predict the age group and gender of hate speech authors. The results show that the answer depends on which demographic attribute is being predicted. This chapter discusses why the task-dependent pattern observed is likely a structural consequence of what hate speech text contains, how the properties of each model type interact with the available signal, and what this implies for cross-domain demographic prediction.

7.1 A Tale of Two Tasks

As reviewed in Chapter 2, author profiling was established as a stylometric task by Koppel et al. (2002) and Schler et al. (2006), both working in relatively uncontrolled corpora where personal style was free to vary. The implicit assumption behind this line of work is that demographic signals are stable across domains. This study puts that assumption under pressure by introducing a domain hate speech in which the communicative register is tightly constrained.

The results reveal a task-dependent pattern. For gender, LLaMA-3.1-8b (macro F1 = 0.515) outperforms all three fine-tuned encoders (best: RoBERTa, 0.488). For age, BERT (macro F1 = 0.310) outperforms both LLMs (best: LLaMA, 0.211). This pattern holds across both LLMs and all three encoders and is therefore unlikely to be model-specific: Qwen3-32B leads HateBERT on gender (0.448 vs. 0.418) but falls below all encoders on age. It should be noted that absolute performance on both tasks remains limited the best gender macro F1 of 0.515 is only modestly above the majority-class baseline of 0.418, and the best age macro F1 of 0.310 is well below a level that would support reliable demographic inference in practice.

One potential explanation for the difference between tasks lies in what each model type brings to the classification. LLMs such as LLaMA-3.1-8b and Qwen3-32B are instruction-tuned on large, diverse text corpora (Ouyang et al., 2022; Touvron et al.,

2023). Through this pretraining, they may have internalized broad demographic associations documented in the prior literature patterns such as hedging and relational language in female-authored text, and more direct, assertive constructions in male-authored text, as reviewed in Chapter 2. Whether such associations transfer to the hate speech domain is not directly testable here, but for the coarser binary gender distinction they appear sufficient to match and slightly exceed fine-tuned encoders trained on roughly 420 authors. As reviewed in Chapter 2, Burger et al. (2011) showed that gender is classifiable from lexical features alone in social media text, supporting the expectation that LLMs would encode related patterns from their pretraining data.

Four-class age prediction requires finer-grained discrimination, and the 66– class presents a specific challenge. One potential explanation is that authors aged 66 and older are substantially underrepresented in the text corpora on which these models are pretrained, limiting their exposure to the language patterns associated with this age group. Additionally, Rangel et al. (2014) observed that the oldest age group was consistently the hardest to classify even under in-domain conditions in the PAN shared tasks; however, the authors do not provide an explicit explanation for this finding, leaving open whether it reflects similar data sparsity or other factors. In the experiments reported here, LLMs do not reliably produce 66– predictions across the five prompt variants tested in this study (see Chapter 4). BERT achieves the highest 66– recall among the five models, and only because the PAN14 training set, after augmentation with additional texts from older authors (described in Chapter 3), contains a non-trivial proportion of 66– examples. A potential reason for the LLM failure is that without labelled hate speech examples from 66– authors, models may be unable to map whatever general patterns they encode about older adult writing to this specific label and domain. Investigating this further for instance by examining whether few-shot examples of 66– authors improve LLM recall remains a direction for future work.

7.2 When the Domain Erases Demographic Signal

Both the LLM advantage on gender and the encoder advantage on age may be partially explained by the same underlying challenge: the hate speech register likely suppresses demographic stylistic variation broadly, though the way this affects gender and age prediction appears to differ. As argued in Chapter 2, Juola (2008) established that genre and register exert a stronger influence on surface linguistic features than individual authorial style. Nini (2019) makes a complementary point: the relationship between linguistic features and demographic variables is itself register-dependent, and stylometric findings established in one register cannot be assumed to transfer to another. In a corpus such as LiLaH, where authors write about similar targets in a politically charged, confrontational register, the genre-level constraints dehumanising framing,

outgroup targeting, and topically constrained vocabulary likely suppress much of the individual demographic variation that classifiers depend on, leaving demographic signal that may be too weak to reliably separate age and gender classes.

Wegmann et al. (2022) describe this as the content–style confound: stylometric classifiers in topically homogeneous corpora tend to learn topic proxies rather than genuine demographic style. The results obtained in this study are consistent with this account. As shown in Figures 7.1 and 7.2 and Tables 7.1 and 7.2, a TF-IDF analysis of the 619 English LiLaH-HAG texts reveals that the vocabulary separating age groups is largely political and immigration-related *europe*, *refugees*, *migrants*, *bbc* rather than stylistic. The gender vocabulary distribution shows a similar pattern: both male and female authors write primarily about *people*, *europe*, *refugees*, and *migrants*, with only a thin margin of male-distinctive aggression (*stupid*, *disgusting*) and female-distinctive policy framing (*borders*, *eu*). This is consistent with Daelemans's (2013) argument that apparently discriminative stylometric features may reflect contextual topic choice rather than stable individual style.

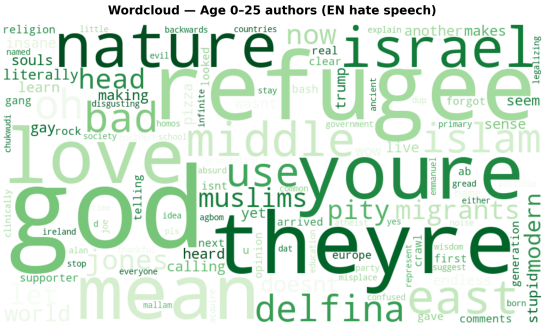


Male authors Female authors

Figure 7.1: Word clouds by gender. Size reflects mean TF-IDF weight.

Table 7.1: Most distinctive vocabulary by gender (TF-IDF difference score).

Male-distinctive	**Female-distinctive**
ur, bigot, says, stupid, hell,	country, countries, eu, women,
world, dont, need, disgusting, use	make, think, day, showing, borders



0–25 26–35

36–65 66–

Figure 7.2: Word clouds by age group. Size reflects mean TF-IDF weight.

Table 7.2: Most characteristic vocabulary by age group (mean TF-IDF weight).

0–25	**26–35**	**36–65**	**66–**
god	people	europe	countries
literally	bigot	people	europe
nature	ur	don	people
just	just	want	country
refugees	like	just	don
heard	marriage	refugees	better
oh	europe	countries	taking
trump	need	like	migrants
mental	know	eu	problems
modern	gay	country	bbc

The two middle age classes 26–35 and 36–65 share a Jaccard vocabulary similarity of 0.439 in LiLaH-HAG and 0.896 in PAN14. In the training corpus, their top vocabularies are almost entirely shared; in the test corpus, they remain the most similar pair by a substantial margin. This structural overlap is consistent with the bidirectional 40–41% confusion between them documented in Chapter 6: a model trained on data where these two groups share nearly all of their most frequent vocabulary may have limited

basis for reliably separating them, regardless of domain. The difficulty with the 66–class appears to stem from a different cause: not vocabulary overlap, but likely training scarcity and the absence of a distinctive surface signature that clearly survives domain transfer.

The content–style confound may also help explain the LLMs' most puzzling error pattern: over-predicting the 0–25 class (232 predictions vs. 34 true). Hate speech comments are often short, terse, and emotionally charged. These surface features resemble the characteristics associated with young authors in PAN14 blog data, which LLMs may have internalized during pretraining abbreviations, minimal punctuation, blunt affect. But in LiLaH, these features reflect the *genre* rather than the *author*: authors of very different age groups writing hate speech comments can produce texts that are, at the surface level, difficult to distinguish as illustrated by the age false-positive examples in Chapter 6 (Section 6.3), where texts from 26–35 and 36–65 authors display the same abbreviated insult register associated with 0–25 authors in the training data.

The cross-domain text length gap reinforces both failures. LiLaH-HAG comments average 41.8 words; PAN14 training chunks average 261 words a sixfold difference. Short texts provide less stylistic surface for models to work with, and the length effect is measurable: age prediction macro F1 for BERT rises from 0.290 on texts of 1–15 words to 0.352 on texts of 61 or more words. The divergence between the two model types at longer texts is also revealing: BERT continues to improve with length, while LLaMA's performance drops from 0.268 to 0.197 in the 61+ bucket, consistent with the interpretation that longer texts tend to contain more topically dense political content, in which any stylistic age signal may be dominated by topic-level variation. The abbreviation case shows the same domain-transfer failure with particular precision: 0–25 authors in PAN14 use abbreviations at 8.2 times the rate of 36–65 authors (mean 0.275 vs. 0.034, $p < 0.001$), giving BERT a genuine learned signal. In LiLaH, however, abbreviated spelling is a feature of the hate speech register rather than of age: the same forms (*ur*, *u*, *ppl*) appear across all age groups, substantially reducing the discriminative value of the abbreviation cue. The signal that was reliable in the source domain appears to lose much of its discriminative value in the target domain.

7.3 The Role of Morphological Cues: Evidence from Slovene

The Slovene experiment tests one specific question: when unambiguous grammatical gender cues are present in the text, does the performance gap between LLMs and encoders narrow? In Slovenian, past-tense verb forms directly encode the speaker's

gender in first-person constructions, providing a surface signal that requires no stylistic inference. Škrjanec et al. (2017) demonstrated that this morphological signal enables reliable gender classification from Slovenian social media text, achieving 92.6% accuracy. Gender classification in morphologically rich languages has been shown to benefit from such grammatical cues more broadly (Markov et al., 2017).

The results partially confirm this expectation. In English, LLaMA-3.1-8b achieves a gender macro F1 of 0.515, compared to 0.488 for the best encoder (RoBERTa) a gap of 2.7 macro F1 points in favour of the LLM. In Slovene, CroSloEngual BERT achieves 0.609, compared to 0.603 for the best Slovene LLM (Qwen3-32B) a gap of only 0.6 macro F1 points in favour of the encoder. Note that the best-performing LLM differs across languages LLaMA-3.1-8b in English and Qwen3-32B in Slovene so this comparison reflects the best available LLM in each setting rather than the same model across languages. The narrowing of this gap in Slovene suggests that the availability of stronger grammatical gender cues in the text may partially explain the observed LLM advantage in English: when such cues are absent, as in English hate speech, a model's pretraining associations may give it a slight edge over an encoder trained on a limited number of labelled authors.

Qwen3-32B's gender predictions also shift substantially across languages. In English, the model produces near-total male predictions (female F1 = 0.065). In Slovene, directed explicitly by the prompt to attend to morphological verb forms, the model produces female predictions more frequently (female F1 = 0.462). This difference is consistent with the interpretation that the male prediction bias in English reflects the limited and noisy gender signal in hate speech text, though the Slovene results remain modest in absolute terms and should not be taken as evidence that the problem is resolved when grammatical cues are present.

The Slovene findings thus partially explain the poor English performance: the absence of grammatically encoded gender cues in English hate speech likely contributes to the difficulty of the task for all model types. Other factors including the cross-domain gap between PAN14 training data and LiLaH-HAG hate speech, and the topically homogeneous register of hate speech may further suppress demographic signal, and could be explored in more detail in future work.

7.4 Limitations

Cross-dataset domain shift. Encoder models were trained on PAN14 English blog text and evaluated on LiLaH-HAG hate speech. The performance gap likely reflects a combination of model capability and domain shift (Pan and Yang, 2010); results may not generalise to in-domain encoder training.

Test set size. With only 64 true 66– examples and 174 female examples in the

619-item test set, per-class F1 estimates for minority groups carry high variance. The BERT 66– F1 of 0.294, for instance, rests on 21 correct predictions.

Slovene encoder comparability. CroSloEngual BERT was trained on Janes-Blog (Slovene blogs, morphology-derived labels), not PAN14. The Slovene results test a different type of cross-dataset transfer and are not directly comparable to the English encoder results.

LLM inference conditions. All LLM results reflect a fixed temperature (0.1), a 1,500-character input truncation limit, and the Groq API. Results characterise these models under these specific conditions.

Annotation reliability. Gender and age labels in LiLaH-HAG are derived from author profile information on social media platforms rather than self-reported demographic identity. The reliability of these labels as proxies for actual demographic attributes is not assessed in this study, and label noise could affect the validity of performance estimates.

Chapter 8

Conclusion

8.1 Summary of Findings

This thesis investigated how well zero-shot large language models and fine-tuned encoder models can predict the age group and gender of hate speech authors, using the English portion of LiLaH-HAG as the evaluation set.

Addressing SQ1, zero-shot LLMs achieve a best binary gender macro F1 of 0.515 (LLaMA-3.1-8b), modestly above the majority-class baseline of 0.418, and a best four-class age macro F1 of 0.211 (LLaMA-3.1-8b), below the uniform random baseline of 0.25. Addressing SQ2, fine-tuned encoders achieve a best gender macro F1 of 0.488 (RoBERTa) and a best age macro F1 of 0.310 (BERT). Addressing SQ3, the best-performing approach differs by task: LLMs produce higher gender macro F1 than all three encoders, while encoders produce higher age macro F1 than both LLMs. This task-dependent pattern is consistent across both LLMs and all three encoders, and is therefore unlikely to be a model-specific effect. It should be noted that absolute performance on both tasks remains limited, and neither approach achieves levels that would support reliable demographic inference in practice. In answer to the main research question: neither approach is universally superior. LLMs perform marginally better on binary gender prediction, while fine-tuned encoders perform better on four-class age prediction, and the margins on both tasks are modest.

A particularly consistent finding concerns the 66– age class. Four of five models do not predict this class for any test example. BERT the only model with training examples from this group achieves a per-class F1 of 0.294, resting on 21 correct predictions out of 64. Across five targeted prompt engineering variants, LLMs do not reliably produce 66– predictions, suggesting this may reflect a structural limitation related to the underrepresentation of this group in pretraining and training corpora rather than a simple artefact of prompt formatting.

The content–style confound (Wegmann et al., 2022) the tendency for classifiers in

topically homogeneous corpora to learn topic proxies rather than genuine demographic style partially accounts for the observed patterns. The vocabulary analysis of LiLaH-HAG (Figures 7.1 and 7.2) shows that the most characteristic vocabulary for each demographic group consists primarily of political and immigration terms that appear across all groups, rather than stylistic markers that reliably distinguish individuals. This is one contributing factor; the cross-domain gap between PAN14 blog training data and LiLaH-HAG hate speech, and the limited number of labelled training authors, are additional factors.

The Slovene experiment offers partial evidence that signal availability contributes to the English gender results. When unambiguous grammatical gender cues are present in the text, the gap between the best LLM (Qwen3-32B, macro F1 = 0.603) and the best encoder (CroSloEngual BERT, macro F1 = 0.609) narrows to 0.6 macro F1 points, compared to 2.7 points in English (LLaMA-3.1-8b: 0.515; RoBERTa: 0.488). Note that different LLMs perform best in each language (LLaMA-3.1-8b in English; Qwen3-32B in Slovene), so this comparison reflects the best available LLM in each setting. These findings are specific to the languages and evaluation set used; whether they generalise to other languages and domains remains an open question.

The prediction patterns documented particularly the consistent underperformance on the 66– age class and the tendency of LLMs to over-predict the male class suggest that caution is warranted when using these models for demographic annotation of hate speech data, as such patterns risk being reproduced in any synthetically annotated training set.

8.2 Future Work

Several directions follow directly from this study.

In-domain encoder training. Training an encoder directly on LiLaH-HAG with cross-validation, rather than on PAN14, would remove the domain shift and establish a cleaner upper bound on what supervised classification can achieve in this corpus.

Extending to other languages. LiLaH-HAG covers Croatian, Dutch, and Slovenian in addition to English. Extending the full evaluation to these languages would test whether the observed task-dependent LLM–encoder differences generalise across typologically distinct settings.

Beyond binary gender. LiLaH-HAG annotations use binary gender labels derived from social media profiles. Future work could examine non-binary or author-reported gender identities, which would require different annotation schemes and modelling approaches.

Few-shot and in-context learning. All LLM experiments in this study are zero-shot. Providing a small number of labelled examples from the target domain as

in-context demonstrations including examples from the 66– group could be a direction for improving LLM performance, particularly on underrepresented classes.

Bibliography

E. M. Bender and B. Friedman. Data statements for natural language processing: Toward best practices. *Transactions of the Association for Computational Linguistics*, 6:587–604, 2018.

J. D. Burger, J. Henderson, G. Kim, and G. Zarrella. Discriminating gender on Twitter. In *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing*, pages 1301–1309, 2011.

T. Caselli, V. Basile, J. Mitrović, and M. Granitzer. HateBERT: Retraining BERT for abusive language detection in English. In *Proceedings of the 5th Workshop on Online Abuse and Harms*, pages 17–25, 2021.

W. Daelemans. Explanation in computational stylometry. In A. Gelbukh, editor, *Computational Linguistics and Intelligent Text Processing, 14th International Conference, CICLing 2013*, pages 451–462. Springer, 2013.

J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

M. Doğanç and I. Markov. From generic to personalized: Investigating strategies for generating targeted counter narratives against hate speech. In *Proceedings of the 1st Workshop on CounterSpeech for Online Abuse (CS4OA)*, pages 1–12, 2023.

D. Fišer, T. Erjavec, and N. Ljubešić. The Janes corpus of Slovenian user-generated content. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*, pages 2428–2433, 2016.

A. Grattafiori, A. Dubey, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

L. Hilte, I. Markov, N. Ljubešić, D. Fišer, and W. Daelemans. Who are the haters? A corpus-based demographic analysis of authors of hate speech. *Frontiers in Artificial Intelligence*, 6, 2023.

X. Huang, L. Xing, F. Dernoncourt, and M. J. Paul. Multilingual Twitter corpus and baselines for evaluating demographic bias in hate speech recognition. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 1440–1448, 2020.

P. Juola. Authorship attribution. *Foundations and Trends in Information Retrieval*, 1 (3):233–334, 2008. doi: 10.1561/1500000005.

M. Koppel, S. Argamon, and A. R. Shimoni. Automatically categorizing written texts by author gender. In *Literary and Linguistic Computing*, volume 17, pages 401–412, 2002.

Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov. RoBERTa: A robustly optimised BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

I. Markov, H. Gómez-Adorno, G. Sidorov, and A. Gelbukh. The winning approach to cross-genre gender identification in Russian at RUSProfiling 2017. In *FIRE 2017 Working Notes*, pages 20–24, 2017.

I. Markov et al. LiLaH: A multilingual hate speech dataset for Linguistic analysis. In *Proceedings of the Language Resources and Evaluation Conference*, 2022.

P. Mishra, M. Del Tredici, H. Yannakoudakis, and E. Shutova. Author profiling for hate speech detection. In *Proceedings of the 27th International Conference on Computational Linguistics*, 2018.

D. Nguyen, R. Gravel, D. Trieschnigg, and T. Meder. "How old do you think I am?": A study of language and age in Twitter. In *Proceedings of the 7th International AAAI Conference on Weblogs and Social Media*, pages 439–448, 2013.

A. Nini. Developing forensic authorship profiling. *Language and Law / Linguagem e Direito*, 5(2):38–58, 2019.

L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35, 2022.

S. J. Pan and Q. Yang. A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10):1345–1359, 2010.

Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

F. Rangel, P. Rosso, I. Chugur, M. Potthast, M. Trenkmann, B. Stein, B. Verhoeven, and W. Daelemans. Overview of the author profiling task at PAN 2013. In *CLEF 2013 Evaluation Labs and Workshop*, 2013.

F. Rangel, P. Rosso, M. Koppel, E. Stamatatos, and G. Inches. Overview of the author profiling task at PAN 2014. In *CLEF 2014 Evaluation Labs and Workshop*, 2014.

F. Rangel, P. Rosso, M. Potthast, B. Stein, and W. Daelemans. Overview of the author profiling task at PAN 2015. In *CLEF 2015 Evaluation Labs and Workshop*, 2015.

F. Rangel, P. Rosso, B. Verhoeven, W. Daelemans, M. Potthast, and B. Stein. Overview of the author profiling task at PAN 2016. In *CLEF 2016 Evaluation Labs and Workshop*, 2016.

J. Schler, M. Koppel, S. Argamon, and J. Pennebaker. Effects of age and gender on blogging. In *AAAI Spring Symposium on Computational Approaches to Analysing Weblogs*, 2006.

I. Škrjanec, D. Fišer, and S. Pollak. Gender profiling for Slovene Twitter communication. In *Proceedings of the 6th Workshop on Balto-Slavic Natural Language Processing*, pages 119–123, 2017.

H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, et al. LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.

M. Ulčar and M. Robnik-Šikonja. FinEst BERT and CroSloEngual BERT. In *Proceedings of the International Conference on Text, Speech, and Dialogue*, pages 104–111, 2020.

A. Wegmann, M. Schraagen, and D. Nguyen. Same author or just same topic? Towards content-independent style representations. In *Proceedings of the 7th Workshop on Representation Learning for NLP*, pages 249–268, 2022.

C. Ziems, W. Held, O. Shaikh, J. Chen, Z. Zhang, and D. Yang. Can large language models transform computational social science? *Computational Linguistics*, 50(1): 237–291, 2024.

Appendix A

Prompt Templates

A.1 Main Prompt Variants

Three prompt variants were tested in preliminary experiments before selecting the best configuration for the full run. All prompts use joint prediction (gender and age in a single call). The *demographic* prompt was selected as best and used for all final LLM experiments.

Baseline Prompt

```
You are an expert in linguistic profiling.

Predict the gender and age group of the author of the following text.

Allowed labels:

Gender:
M
F

Age:
0-25
26-35
36-65
66-

Rules:
- Choose exactly one gender and one age group.
- If uncertain, choose the most likely option.
```

```
- Output only in the required format, nothing else.

Output format:
Gender: <M or F>
Age: <0-25 | 26-35 | 36-65 | 66->

Text:
"""
{TEXT}
"""
```

Style Prompt

```
You are a linguistic analyst specialized in author profiling.

Determine the author's gender and age group based ONLY on writing style cues.

Focus on:
- vocabulary and word choice
- use of slang or formal language
- punctuation and capitalization
- sentence length and structure
- tone and emotional expression
- degree of formality

Allowed labels:

Gender:
M
F

Age:
0-25
26-35
36-65
66-

Rules:
- Choose exactly one gender and one age group.
```

```
- If uncertain, choose the most likely option.
- Do not explain your reasoning.
- Output only in the required format, nothing else.

Output format:
Gender: <M or F>
Age: <0-25 | 26-35 | 36-65 | 66->

Text:
"""
{TEXT}
"""
```

Demographic Prompt (selected)

```
You are a linguistic analyst specialized in demographic author profiling.

Predict the gender and age group of the author based on linguistic patterns.

Demographic markers to consider:

Gender:
- M: direct assertions, less hedging, more aggressive or blunt language,
     fewer qualifiers
- F: hedging phrases ("I think", "maybe", "perhaps"), more emotional
     expression, more qualifiers and politeness markers

Age:
- 0-25:  casual language, internet slang, abbreviations, short sentences,
         informal tone, minimal punctuation
- 26-35: mix of formal and informal, complete sentences, references to
         work or adult life
- 36-65: more formal vocabulary, longer sentences, traditional expressions,
         less slang
- 66-:   very formal language, old-fashioned expressions, avoids slang,
         longer and more complex sentences

Allowed labels:
```

Gender:
M
F

Age:
0-25
26-35
36-65
66-

Rules:
- Choose exactly one gender and one age group.
- If uncertain, choose the most likely option.
- Do not explain your reasoning.
- Output only in the required format, nothing else.

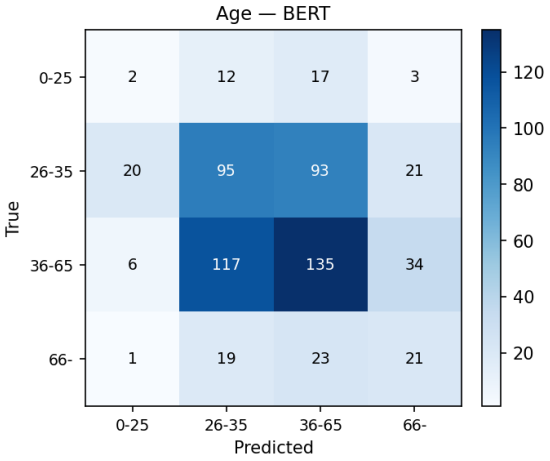
Output format:
Gender: <M or F>
Age: <0-25 | 26-35 | 36-65 | 66->

Text:
"""
{TEXT}
"""

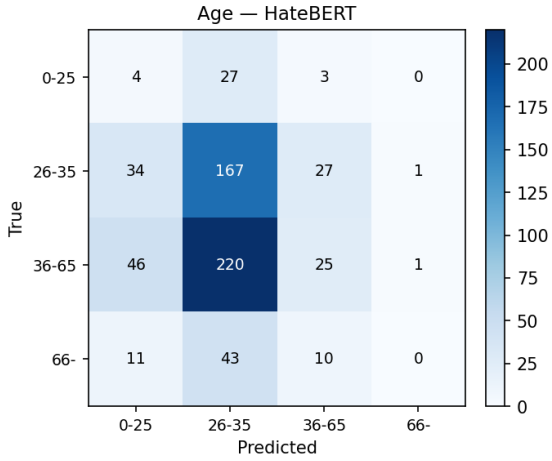
Appendix B

Full Confusion Matrices

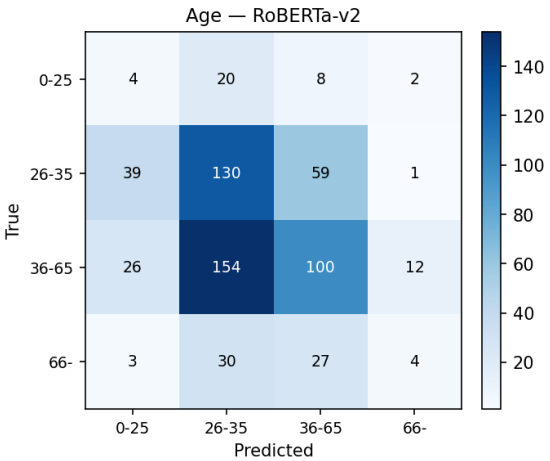
B.1 Age Prediction — All Models



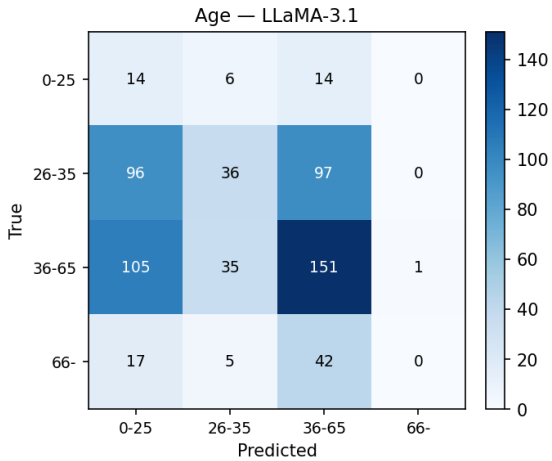
(a) BERT



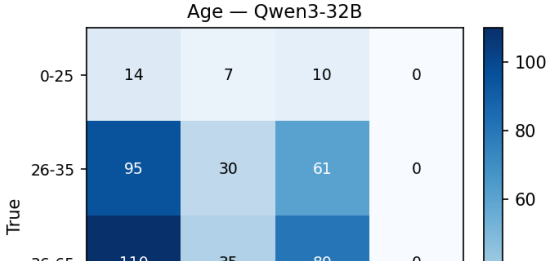
(b) HateBERT



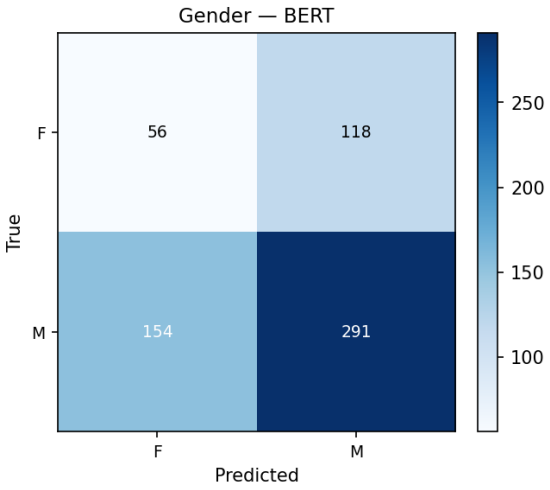
(c) RoBERTa



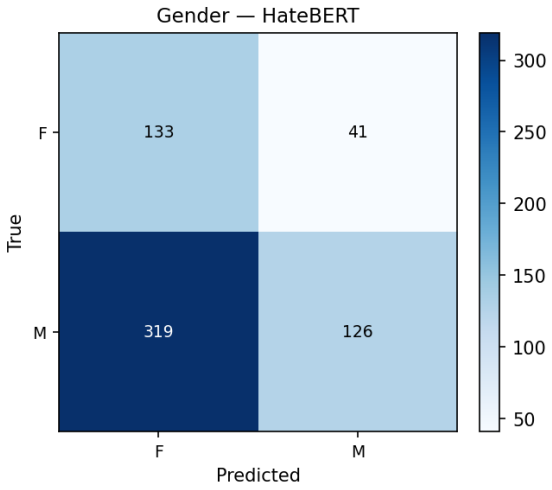
(d) LLaMA-3.1-8b



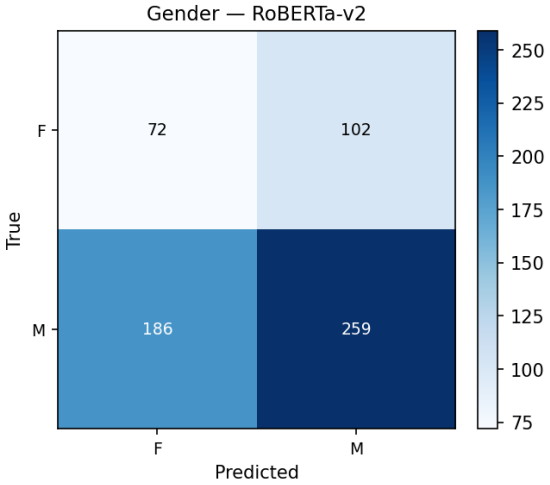
B.2 Gender Identification — All Models



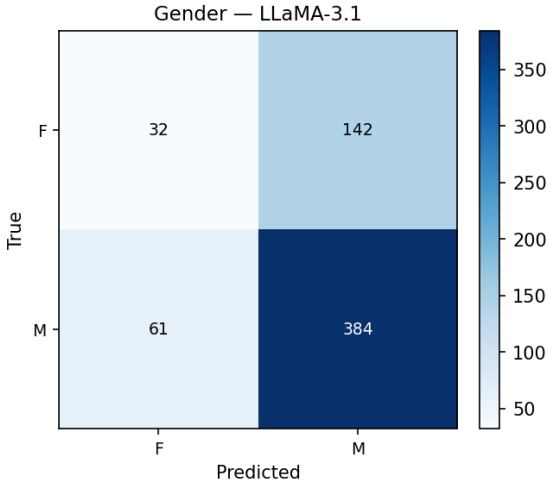
(a) BERT



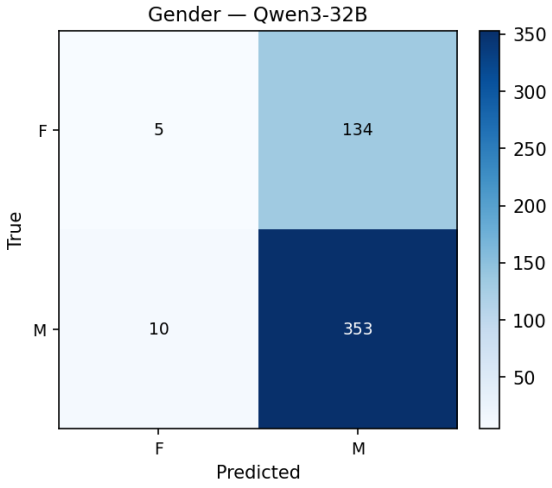
(b) HateBERT



(c) RoBERTa



(d) LLaMA-3.1-8b



(e) Qwen3-32B

Figure B.2: Gender identification confusion matrices for all five models.

Appendix C

Data Statements

The following data statements follow the framework proposed by Bender and Friedman (2018). A data statement is a structured characterisation of a dataset that documents who produced the texts, under what conditions, with what annotations, and with what potential limitations. Two data statements are provided: one for LiLaH-HAG (the evaluation set) and one for the PAN14 Author Profiling corpus (the encoder training set).

C.1 LiLaH-HAG

A. Curation Rationale
LiLaH-HAG (Markov et al., 2022) was collected to support research on hate speech detection and author profiling in a multilingual setting. Texts were drawn from public Facebook comment sections associated with politically charged topics, primarily related to immigration and refugees in Europe. Selection was guided by the presence of hate speech content and the availability of author profile information sufficient to derive demographic labels. Only comments for which complete age group and gender labels could be assigned were retained. This thesis uses only the English-language portion of the dataset.

B. Language Variety
The full dataset covers four languages: English (EN), Croatian (HR), Dutch (NL), and Slovenian (SL). All four varieties are written, informal, and platform-specific (Facebook comment register). This study uses the English subset exclusively ($n = 619$ after filtering for complete annotations).

C. Speaker Demographics
Authors are Facebook users who posted publicly on pages related to immigration and refugee topics. Demographic information was inferred from public Facebook profile data rather than collected through self-report.

- **Age**: Four groups based on author year of birth. Distribution in the English subset: 0–25 (5.5%, $n = 34$), 26–35 (37.0%, $n = 229$), 36–65 (47.2%, $n = 292$), 66– (10.3%, $n = 64$). Age labels reflect the author's age at the time of posting.
- **Gender**: Binary (Male / Female), derived from Facebook profile information. Distribution: Male 71.9% ($n = 445$), Female 28.1% ($n = 174$). Labels do not capture non-binary or fluid gender identities.
- **Native language and socioeconomic background**: Not documented in the dataset.

D. Annotator Demographics

Hate speech content labels (target group, severity) were assigned by trained human annotators following guidelines described in Markov et al. (2022). Demographic labels (age, gender) were not assigned by human annotators: they were derived algorithmically from author Facebook profile metadata. No annotator demographic information is reported for the profile-derived labels. Label reliability for the demographic attributes is not assessed in the original dataset or in this thesis.

E. Speech Situation

Texts are written, asynchronous, public-facing comments posted on Facebook pages. Authors were aware their posts were publicly visible but may not have anticipated use in research. The communicative situation is adversarial and confrontational: comments target outgroups (immigrants, refugees, specific nationalities or religious groups) with dehumanising or aggressive language. Texts are produced without an expectation of formal register.

F. Text Characteristics

Short, informal hate speech comments. Mean length 41.8 words (median 26 words, range 1–392 words). Register is consistently confrontational and politically charged across all demographic groups, with frequent use of abbreviated spelling, profanity, direct address, and topically constrained vocabulary (immigration, religion, national identity). Stylistic variation attributable to individual demographic characteristics is limited by the tight topical and register constraints of the domain.

G. Provenance Appendix

Source: Publicly available Facebook comment sections. **Citation**: Markov et al. (2022). **Access**: The dataset is available from the original authors upon request; it is not redistributed in this thesis. The English portion used here consists of 619 examples. No raw Facebook data is reproduced in this thesis; all quoted examples are presented solely for research illustration.

C.2 PAN14 Author Profiling Corpus

A. Curation Rationale

The PAN14 Author Profiling corpus (Rangel et al., 2014) was created for the PAN 2014 shared task on author profiling, with the goal of benchmarking automatic prediction of author age and gender from social media text. The English portion consists of blog posts whose authors provided demographic information through platform profiles or self-disclosure. The dataset was used in this thesis as the training source for fine-tuned encoder models. Preprocessing included chunking long author documents into shorter passages to match transformer input constraints, and augmenting the 66– age group with additional review texts by the same authors to mitigate severe class imbalance (see Chapter 3).

B. Language Variety

The original PAN14 dataset covers English and Spanish; this thesis uses the English portion only. Texts are blog posts written in standard written English, with register ranging from informal personal diaries to semi-formal opinion writing. The variety is substantially more discursive and longer-form than the LiLaH-HAG register.

C. Speaker Demographics

Authors are English-speaking bloggers drawn from general blogging platforms.

- **Age**: Four groups. Distribution after preprocessing and augmentation (n = 11,906 chunks): 0–25 (16.0%, n = 1,910), 26–35 (26.0%, n = 3,098), 36–65 (51.1%, n = 6,084), 66– (6.8%, n = 814). The 66– group was augmented from an initial representation of approximately 0.2% of text chunks. All chunks from a given author share the same demographic label.
- **Gender**: Binary (Male / Female). Distribution is balanced at the author level (50% / 50%) as a result of the shared task sampling strategy; this balance is preserved in the preprocessed training set.
- **Native language and socioeconomic background**: Not documented.

D. Annotator Demographics

Age and gender labels were collected through author self-report via platform profile information. No human annotation of demographic attributes was performed. Annotator demographic information is therefore not applicable for these labels. The shared task organisers are responsible for data collection and label verification; details are provided in Rangel et al. (2014).

E. Speech Situation

Texts are written blog posts, produced asynchronously for a general public audience.

Authors self-selected to write publicly on blogging platforms and consented to public visibility of their posts. The communicative situation is personal, opinionated, or diary-style writing, substantially different from the adversarial, confrontational register of LiLaH-HAG. This register difference is the primary source of domain shift in the cross-dataset evaluation.

F. Text Characteristics

Long-form blog posts chunked to training-length passages. Mean chunk length approximately 261 words — roughly six times the mean length of LiLaH comments. Register is more varied and reflective than LiLaH. Topics include personal life, opinions, and general interest writing, without the topically homogeneous political content that characterises LiLaH.

G. Provenance Appendix

Source: PAN 2014 Author Profiling shared task, hosted through CLEF. **Citation**: Rangel et al. (2014). **Access**: Available through the PAN/CLEF infrastructure to registered participants. Augmentation texts consist of additional review texts from the same 66– authors, sourced from a publicly accessible review corpus. Augmented training data is not redistributed in this thesis.